The Law of Portfolio Learnability

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I. Introduction

In the Markowitz (1952) framework, first the portfolio manager estimate the two moments of returns, secondly the PM estimate the portoflio wieghts optimising a criterion.

A different view emerged with the direct portfolio approach of Brandt (1999) and Brandt et al. (2009). Investors ultimately care about portfolio weights, not forecasts of moments. Rather than estimating conditional expected returns and covariance matrices as intermediate objects, one can parameterize and estimate the portfolio policy directly. This changed the object of estimation. This was a very futuristic approach, anticipating of years what computer science and machine learning is doing now.

This estimation problem becomes particularly acute in high-dimensional finance. Kozak et al. (2020) show that the estimation becomes unstable in low-eigenvalue directions. Where the eigenvalues tend to zero, the payoff of the eigenportfolio tend to infinite, creating an arbitrage opportunity. This motivate the economically shrinkage.

More recently, the machine-learning literature has brought model complexity to the foreground. Empirical asset pricing means model with data the financial market. There is always a model behind every modelization. Thus finance can't ignore what is happening in the machine leanring world. Indeed Kelly et al. (2024) and Kelly and Malamud (2025) show that increasing model complexity can improve out-of-sample return prediction even deep into the overparameterized regime. Their conclusion is deliberately strong:

> *"The analyst should always use the largest approximating model that she can compute."*

This was strongly new for the finance world in which the only virute was the parsimony.

This "virtue of complexity" shows that overparameterization can improve approximation and generate useful implicit regularization. Nagel (2025), in turn, shows why the economic meaning of highly overparameterized estimators must be interpreted carefully: large random-feature models

can converge to kernel rules whose behavior is driven by particular forms of historical weighting rather than by an arbitrarily rich set of distinct economic decisions, introducing nominally and the effective complexity.

The overparatized model not only now are in returns prediction, but appears in directly portfolio choice. Simon et al. (2026) replace the linear parametric portfolio policy with deep neural networks and show that flexible nonlinear policies can generate economically valuable allocations, while investor preferences themselves act as a form of economic regularization.

Taken together, this literature establishes that rich representations can be useful and that regularization is if not indispensable, at least usefull.

Yet several fundamental questions remain open.

Building on the direct portfolio approach of Brandt: How reliably, and how quickly, can a flexible and "complex" portfolio rule recover the population-optimal investment decision?
Building on the spectral shrinkage argument of Kozak, Nagel, and Santosh: Which investment directions should be shrunk because they are only weakly identified by returns?
Building on the virtue-of-complexity literature: When is a richer representation a virtue for portfolio choice, and what notion of complexity is economically relevant for the investment decision itself?

Answering these questions requires starting from a more fundamental one:

How much of the optimal portfolio can be learned from a finite market history?

Answering this question requires establishing whether and how fast the portfolio decision can be learned directly, identifying which investment directions should be shrunk because they are only weakly revealed by realized returns, and determining what notion of complexity is economically relevant for portfolio choice—and how much of that complexity a finite market history can support.

First we used a particular estimator (Kernel ridge regression) to quantify our results, later we generalize these to every estimator. Our first contribution is to quantify how rapidly direct portfolio estimation recovers the population tangency direction as market history accumulates. We prove that the directly weights estimation for maximizing the Sharpe ratio is consistent and we characterize the rate at which its estimated direction converges to the population Sharpe. We find that portfolio learnability is governed by the spectrum of managed portfolios. Directions associated with large eigenvalues correspond to investment opportunities that generate substantial payoff variation and are therefore well identified from the available return history. By contrast, low-eigenvalue directions are only weakly identified, so taking large positions along them makes the portfolio especially sensitive to estimation error. More available portfolio directions therefore do not necessarily translate into more usable investment opportunities: what matters is how much of the opportunity set is concentrated in directions that can be estimated with sufficient precision. The result therefore provides both a formal justification for direct portfolio estimation and a quantitative answer to how much market history is required to learn the optimal investment decision.
Our first contribution implies even a quantification of the optimal shrinkage: It should decline as market history accumulates and as a function of the managed portfolio spectrum. Additional observations progressively reveal portfolio directions that were previously too weakly identified to support reliable investment positions, allowing the investor to exploit a broader set of economically meaningful opportunities.

Our second contribution is what we call the law of effective portfolio complexity. We distinguish two notions of complexity that play fundamentally different economic roles. The first is nominal representation complexity: the size of the approximating representation, as measured, for example, by the number of features, basis functions, or parameters. Kelly, Malamud, and Zhou show that, under their proportional high-dimensional asymptotic framework, increasing nominal representation size can improve limiting out-of-sample return-prediction performance. Richer representations can therefore be valuable because they expand approximation capacity and the set of investment rules that can potentially be expressed.

The second notion is effective portfolio complexity, which we define directly in the space of the investment decision. It measures the effective breadth of the portfolio: the effective number of economically distinct managed-portfolio directions that remain active after shrinkage. This quantity need not move one-for-one with representation size. A representation may be arbitrarily rich while the portfolio learned from it remains effectively low dimensional. We show that the optimal level of effective portfolio complexity is finite at every finite sample size and increases as additional market history becomes available. Too little effective complexity leaves valuable investment opportunities unused; moving beyond the data-supported scale increasingly activates directions that the available history only weakly identifies. A rich representation may be a virtue; how much of that richness should become an active investment decision is a separate learnability question.

Our third and last contribution is what we call the fundamental limit of portfolio learning. We show that the rate at which the optimal portfolio can be recovered is not merely a property of our estimator. No portfolio-learning procedure —including neural networks, random forests, or alternative kernel methods— can uniformly improve this rate over the same class of economies without imposing additional structure.

The learning limit therefore does not belong to a particular estimator. It belongs to the investment problem itself.

Taken together, our results characterize how rapidly the population tangency portfolio can be recovered from finite market history, how richer representations expand the set of potential investment opportunities, and how the effective portfolio complexity that can be reliably exploited grows as additional data become available. We also establish the fastest rate at which any procedure can achieve this recovery. Our results therefore quantify both the opportunities created by increasingly flexible representations and the fundamental limits that finite market history imposes on machine learning for portfolio choice in finance.

II. Population Problem Portfolio Choice

At date t, the portfolio manager faces N risky stocks. The corresponding next-period excess-return vector is $R_{t+1} = \left(R_{1,t+1}, \ldots, R_{N,t+1}\right)' \in \mathbb{R}^N$. For each stock $i = 1, \ldots, N$, the portfolio manager observes a D-dimensional vector of firm characteristics $Z_{i,t} \in \mathcal{Z} \subseteq \mathbb{R}^D$. The portfolio manager has to choose the portfolio weights policy, between an infinte number of policies in an Hilbert space $\mathcal{H}$, at every time, this maps characteristics into weights: $W \in \mathcal{H} : \mathcal{Z} \to \mathbb{R}$. The resulting portfolio return at each time is

$$R^p_{t+1}(W) = \frac{1}{N} \sum_{i=1}^{N} W(Z_{i,t}) R_{i,t+1}. \tag{1}$$

The factor $1/N$ is a cross-sectional normalization that prevents portfolio scale from growing mechanically with the number of available stocks.

Definition 1 (Economy)**.** An economy ε is a probability law for the stochastic process

$$\{(N, Z_t, R_{t+1})\}_{t \in \mathbb{Z}},$$

defined on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \in \mathbb{Z}}, \mathbb{P}_\varepsilon)$. We require Z_t to be $\mathcal{F}_t$-measurable and R_{t+1} to be $\mathcal{F}_{t+1}$-measurable.

For a fixed economy, expectations are taken under $\mathbb{P}_\varepsilon$; we suppress the economy subscript when it is unambiguous. The population criterion is

$$Q(W) = \mathbb{E}\left[\left(1 - R^p_{t+1}(W)\right)^2\right], \qquad W^\star \in \arg\min_{W \in \mathcal{H}} Q(W). \tag{2}$$

The following proposition relates this criterion to the maximum squared Sharpe ratio. All population statements refer to the same one-period state-return distribution.

Proposition 1 (Maximum-Sharpe direction; cf. Britten-Jones (1999))**.** *Suppose every policy in $\mathcal{H}$ generates a square-integrable payoff, the population minimum is attained, and $0 < Q(W^\star) < 1$. Sharpe ratios below are evaluated only for policies with positive payoff variance. Every nonzero policy $W \in \mathcal{H}$ can be written as*

$$W = cU, \qquad c \in \mathbb{R}, \qquad \|U\|_{\mathcal{H}} = 1.$$

The normalized policy U determines the portfolio direction, namely the relative state-dependent allocation rule, whereas c determines the overall scale of exposure along that direction. Then, with the minimization over U restricted to positive-variance payoffs,

$$\min_{W \in \mathcal{H}} Q(W) = \min_{\substack{U \in \mathcal{H} \\ \|U\|_{\mathcal{H}}=1}} \frac{1}{1 + \mathrm{SR}^2\left(R^p_{t+1}(U)\right)},$$

where

$$\mathrm{SR}\left(R^p_{t+1}(U)\right) := \frac{\mathbb{E}\left[R^p_{t+1}(U)\right]}{\sqrt{\mathrm{Var}\left(R^p_{t+1}(U)\right)}}.$$

Hence, minimizing Q selects a maximum-Sharpe direction and fixes its scale, with positive expected excess return.

We embrace the philosophy of Brandt (1999): rather than estimating expected returns and the covariance matrix as intermediate objects and then solving for the optimal portfolio, we estimate portfolio weights directly. Estimating the mean return vector requires estimating N expected returns, while estimating the covariance matrix requires estimating $N(N+1)/2$ distinct covariance parameters. By contrast, directly estimating the portfolio means estimating the N portfolio weights that are ultimately the object of interest.

The criterion in (2) deliberately describes a frictionless portfolio problem. More generally, the loss should represent the economic objective actually faced by the portfolio manager. Transaction costs,

market impact, short-selling costs, leverage constraints, or other implementation frictions should therefore be incorporated directly into the portfolio objective whenever they are economically relevant. Estimating the frictionless-optimal policy and subtracting costs afterwards generally does not deliver the same decision.

Proposition 2 (Pricing kernel and Hansen–Jagannathan frontier)**.** *Let $W_\varepsilon^\star$ denote the population-optimal policy in Proposition 1, and define*

$$m_{t+1,\varepsilon}^\star := 1 - R_{t+1}^p(W_\varepsilon^\star).$$

Then $m_{t+1,\varepsilon}^\star$ prices every managed excess-return portfolio:

$$\mathbb{E}_\varepsilon\left[m_{t+1,\varepsilon}^\star R_{t+1}^p(V)\right] = 0, \qquad \forall V \in \mathcal{H}.$$

Moreover, letting $\mathrm{SR}_\varepsilon^\star := \mathrm{SR}(R_{t+1}^p(W_\varepsilon^\star))$*,*

$$\mathbb{E}_\varepsilon[m_{t+1,\varepsilon}^\star] = \mathbb{E}_\varepsilon[(m_{t+1,\varepsilon}^\star)^2] = \frac{1}{1+(\mathrm{SR}_\varepsilon^\star)^2}.$$

Hence the normalized pricing kernel

$$\widetilde{m}_{t+1,\varepsilon} := \frac{m_{t+1,\varepsilon}^\star}{\mathbb{E}_\varepsilon[m_{t+1,\varepsilon}^\star]}$$

satisfies $\mathbb{E}_\varepsilon[\widetilde{m}_{t+1,\varepsilon}] = 1$*, and* $\mathrm{Var}_\varepsilon(\widetilde{m}_{t+1,\varepsilon}) = (\mathrm{SR}_\varepsilon^\star)^2$*. Equivalently,*

$$\frac{\sigma_\varepsilon(\widetilde{m}_{t+1,\varepsilon})}{\mathbb{E}_\varepsilon[\widetilde{m}_{t+1,\varepsilon}]} = \mathrm{SR}_\varepsilon^\star,$$

so the Hansen–Jagannathan bound ((Hansen and Jagannathan, 1991)) holds with equality.

The proposition provides the asset-pricing dual of Proposition 1. The same portfolio that maximizes the Sharpe ratio generates, through $m_{t+1,\varepsilon}^\star = 1 - R_{t+1}^p(W_\varepsilon^\star)$, the minimum-variance unit-mean

pricing kernel for the managed-payoff space. Thus the maximum-Sharpe portfolio and the Hansen–Jagannathan frontier are two representations of the same population geometry.

III. Direct Portfolio Learning

To estimate the portfolio policy we use a kernel representation Filipović and Schneider (2024); Nagel (2025) because it provides a common framework for portfolio policies of very different forms, as shown in table .... Indeed, depending on the kernel, the same estimator can generate static, linear, polynomial or sieve-based, and flexible nonlinear mappings, generalizing respectively the static mean–variance portfolio of Britten-Jones (1999), the linear parametric portfolio policies of Brandt et al. (2009), and the semiparametric spline-based policies of Caldeira et al. (2023).
Kernel methods are also closely connected to the recent work of Nagel (2025); Kelly and Malamud (2025). In particular, random Fourier features (Rahimi and Recht, 2007) provide finite-dimensional approximations to stationary kernels, while the corresponding kernel estimator can be viewed as the infinite-feature limit. In this sense, the kernel formulation provides a natural benchmark for portfolio models whose nominal representation complexity is allowed to become arbitrarily large.

Kernel methods are also closely related to neural-network representations. A random-feature approximation can be interpreted as a neural network with a fixed, randomly generated hidden layer, in which only the final linear coefficients are estimated. More generally, neural tangent kernels arise as the infinite-width limits of particular neural-network training regimes (Jacot et al., 2018).

Let

$$K : \mathscr{Z} \times \mathscr{Z} \to \mathbb{R}$$

be a positive-definite scalar kernel and let $\mathcal{H}_K$ denote the corresponding space of characteristic-based portfolio policies.

Assumption 1 (Portfolio representation)**.** The space $\mathscr{Z}$ is Polish. Fix a jointly measurable kernel K, with separable RKHS $\mathcal{H}_K$, satisfying

$$\sup_{z \in \mathscr{Z}} K(z, z) \leq \kappa_K < \infty.$$

The representation is fixed across economies and sample sizes.

Let $W_K^\star$ denote the minimum-norm population minimizer in $\mathcal{H}_K$, when attained. Write $W_\varepsilon^\star = W_{\varepsilon,K}^\star$ and $\Sigma_{\mathcal{H}_K,\varepsilon}$ when distinguishing economies.

The empirical regularized counterpart of the population problem in Equation (2) is

$$\begin{aligned}\widehat{W}_{\lambda,T} &:= \underset{W\in\mathcal{H}_K}{\arg\min}\left\{\frac{1}{T}\sum_{t=1}^{T}\left(1-\frac{1}{N}\sum_{i=1}^{N}W(Z_{i,t})R_{i,t+1}\right)^2+\lambda\|W\|_{\mathcal{H}_K}^2\right\}\\ &= \underset{W\in\mathcal{H}_K}{\arg\min}\left\{\widehat{Q}(W)+\lambda\|W\|_{\mathcal{H}_K}^2\right\} = \min_{\substack{U\in\mathcal{H}\\ \|U\|_{\mathcal{H}}=1}}\left\{\frac{1}{1+\widehat{\mathrm{SR}}_{\lambda,T}^2\left(R_{t+1}^p(U)\right))}\right\}, \qquad \lambda>0.\end{aligned} \tag{3}$$

where

$$\widehat{\mathrm{SR}}_{\lambda,T}\left(R_{t+1}^p(U)\right) := \frac{\frac{1}{T}\sum_{t=1}^{T}R_{t+1}^p(U)}{\sqrt{\frac{1}{T}\sum_{t=1}^{T}\left(R_{t+1}^p(U)-\frac{1}{T}\sum_{s=1}^{T}R_{s+1}^p(U)\right)^2+\lambda}}.$$

The economic structure of the estimator becomes transparent by viewing kernel-managed portfolio payoffs as elements of the portfolio policy space. At each date t, define

$$X_t := \frac{1}{N}\sum_{i=1}^{N}R_{i,t+1}K(\cdot,Z_{i,t})\in\mathcal{H}_K.$$

Thus X_t is a random element of $\mathcal{H}_K$. If evaluated at $z\in\mathcal{Z}$, it gives

$$X_t(z) = \frac{1}{N}\sum_{i=1}^{N}R_{i,t+1}K(z,Z_{i,t})\in R.$$

the payoff generated by the kernel-managed portfolio that assigns stock i the weight $K(z,Z_{i,t})/N$. Its population mean is

$$\mu_{\mathcal{H}_K} := \mathbb{E}[X_t]\in\mathcal{H}_K,$$

while its population second-moment operator is

$$\Sigma_{\mathcal{H}_K} := \mathbb{E}[X_t \otimes X_t] : \mathcal{H}_K \to \mathcal{H}_K,$$

where, for every $V \in \mathcal{H}_K$,

$$(X_t \otimes X_t)V = \langle V, X_t \rangle_{\mathcal{H}_K} X_t.$$

The first moment captures expected managed-portfolio payoffs, while the second-moment operator describes the magnitude of realized payoffs across portfolio directions.
Their sample counterparts are the element

$$\widehat{\mu}_T = \frac{1}{T} \sum_{t=1}^{T} X_t \in \mathcal{H}_K,$$

and the operator

$$\widehat{\Sigma}_T = \frac{1}{T} \sum_{t=1}^{T} X_t \otimes X_t : \mathcal{H}_K \to \mathcal{H}_K.$$

Proposition 3 (Managed-portfolio Markowitz representation)**.** *The population-optimal minimum-norm policy satisfies*

$$\mu_{\mathcal{H}_K} = \Sigma_{\mathcal{H}_K} W_K^\star. \tag{4}$$

Equivalently,

$$W_K^\star = \Sigma_{\mathcal{H}_K}^\dagger \mu_{\mathcal{H}_K},$$

where $\Sigma_{\mathcal{H}_K}^\dagger$ denotes the pseudo inverse.
The regularized sample policy in Equation (3) *satisfies*

$$\widehat{W}_{\lambda,T} = \left(\widehat{\Sigma}_T + \lambda I\right)^{-1} \widehat{\mu}_T, \qquad \lambda > 0. \tag{5}$$

Equation (5) provides the structural link to classical portfolio choice. The investor is still com-

bining expected returns against second moments, as in Markowitz. The difference is that the investment objects are now characteristic-managed portfolios weighted with the kernels.

Regularization has a familiar portfolio interpretation. As emphasized by Kozak et al. (2020), portfolio directions with very little payoff variation but non-negligible expected returns resemble near-arbitrage opportunities and should therefore be treated with particular caution. Regularization has a familiar portfolio interpretation. As emphasized by Kozak et al. (2020), weak portfolio directions are particularly dangerous because small estimation errors can translate into very large portfolio positions. In their eigenportfolio setting, a low-variance direction with a sufficiently large estimated expected return can even resemble a near-arbitrage opportunity.

The same instability appears in our managed-portfolio representation. Let $\Sigma_{\mathcal{H}_K} e_j = \mu_j e_j$, and let $m_j := \langle \mu_{\mathcal{H}_K}, e_j \rangle_{\mathcal{H}_K}$. Without regularization, the portfolio weight along direction e_j is proportional to

$$\widehat{W}_{0,j} = \frac{\widehat{m}_j}{\widehat{\mu}_j}.$$

Hence, when $\widehat{\mu}_j$ is small, even modest estimation error in $\widehat{m}_j$ can generate a very large portfolio position. Regularization replaces the small eigenvalue in the denominator by $\widehat{\mu}_j + \lambda$:

$$\widehat{W}_{\lambda,j} = \frac{\widehat{m}_j}{\widehat{\mu}_j + \lambda} = \frac{\widehat{\mu}_j}{\widehat{\mu}_j + \lambda} \widehat{W}_{0,j}.$$

Thus, $\widehat{\mu}_j/(\widehat{\mu}_j + \lambda)$ is the spectral shrinkage factor.

The spectral filter derived above suggests a natural measure of how much of the managed-portfolio opportunity set is effectively used by the investor. The relevant notion of portfolio complexity is not the number of characteristics, parameters, or random features in the representation, but the number of economically distinct portfolio directions that remain active after shrinkage.

This motivates the following definition.

Definition 2 (Effective portfolio complexity)**.** For a fixed portfolio representation K and $\lambda > 0$,

define

$$C_K(\lambda) := \mathrm{Tr}\left[\Sigma_{\mathcal{H}_K}(\Sigma_{\mathcal{H}_K} + \lambda I)^{-1}\right] = \sum_{j\geq 1} \frac{\mu_j}{\mu_j + \lambda}. \tag{6}$$

Effective portfolio complexity is the shrinkage-adjusted number of managed-portfolio directions used by the portfolio rule.

The definition follows directly from the spectral shrinkage mechanism already introduced. Each managed-portfolio direction contributes between zero and one to $C_K(\lambda)$. If $\mu_j \gg \lambda$, its shrinkage factor is close to one and the direction is effectively active. If $\mu_j \ll \lambda$, the direction contributes little and is effectively suppressed.

Effective complexity therefore measures how deeply into the managed-portfolio spectrum the investor is actually operating.

This notion is fundamentally different from nominal representation size. A model may contain thousands, or infinitely many, basis directions while the portfolio decision itself remains effectively low dimensional because regularization suppresses most weak directions. Conversely, a rich representation can be valuable because it makes additional investment opportunities available once sufficient market history exists to identify them.

IV. Portfolio Learnability

A. A Class of Portfolio Economies

For the fixed representation in Assumption 1, fix $b > 1$, $\gamma > b$, constants $C_\beta, B_X, R > 0$, $K_4 \geq 1$, and $0 < c_\mu \leq C_\mu < \infty$. We denote by

$$\mathcal{E} = \mathcal{E}(b, \gamma, C_\beta, B_X, K_4, R, c_\mu, C_\mu)$$

the nonempty class of economies ε satisfying the following conditions, with all bounds common to the class:

(E1) The process $\{(Z_t, R_{t+1})\}_{t\in\mathbb{Z}}$ is strictly stationary and polynomially β-mixing, with $\beta_\varepsilon(k) \leq C_\beta k^{-\gamma}$ for every $k \geq 1$.

(E2) *Gaussian-type managed-return tails and conditional fourth moments.* Let

$$K_t := \left[K(Z_{i,t}, Z_{j,t})\right]_{i,j=1}^{N}.$$

Assume

$$\mathbb{E}_\varepsilon\left[\exp\left(\frac{N^{-2}R'_{t+1}K_tR_{t+1}}{B_X^2}\right)\middle|\mathscr{F}_t\right] \leq 2 \quad \text{a.s.} \tag{7}$$

For every $V \in \mathcal{H}_K$, assume also

$$\mathbb{E}_\varepsilon\left[(R^p_{t+1}(V))^4\middle|\mathscr{F}_t\right] \leq K_4\left\{\mathbb{E}_\varepsilon\left[(R^p_{t+1}(V))^2\middle|\mathscr{F}_t\right]\right\}^2 \quad \text{a.s.} \tag{8}$$

Both bounds are uniform over the class. Since $N^{-2}R'_{t+1}K_tR_{t+1} = \|X_t\|^2_{\mathcal{H}_K}$, Condition (7) directly controls the diagonal of the induced kernel while allowing unbounded returns. Conditional Gaussian returns with uniformly bounded conditional second moments satisfy Condition (E2); see Proposition 4.

(E3) The population minimum over $\mathcal{H}_K$ is attained. Its minimum-norm solution $W_\varepsilon^\star$ satisfies $\|W_\varepsilon^\star\|_{\mathcal{H}_K} \le R$.

(E4) The pricing kernel $m_{t+1,\varepsilon}^\star := 1 - R_{t+1}^p(W_\varepsilon^\star)$ satisfies

$$\mathbb{E}_\varepsilon\left[m_{t+1,\varepsilon}^\star R_{t+1}^p(V) \mid \mathscr{F}_t\right] = 0, \qquad \forall\, V \in \mathcal{H}_K, \quad \text{a.s.}$$

Here $\mathscr{F}_t$ is the investor's information at the decision date in Definition 1. Conditional pricing is imposed separately from unconditional population optimality; positivity is not imposed.

(E5) Let $\Sigma_{\mathcal{H}_K,\varepsilon} = \mathbb{E}_\varepsilon[X_t \otimes X_t]$ and let $\mu_{1,\varepsilon} \ge \mu_{2,\varepsilon} \ge \cdots > 0$ denote its positive eigenvalues. Then

$$c_\mu j^{-b} \le \mu_{j,\varepsilon} \le C_\mu j^{-b}, \qquad j \ge 1.$$

(E6) The optimal payoff has positive variance and $\mathrm{SR}_{\varepsilon,K}^\star := \mathrm{SR}(R_{t+1}^p(W_\varepsilon^\star))$ satisfies the common bounds

$$0 < \underline{S} \le \mathrm{SR}_{\varepsilon,K}^\star \le \overline{S} < \infty.$$

Condition (E2) replaces the almost-sure bounded-return restriction by Gaussian-compatible tail and fourth-moment control. Proposition 4 in Appendix A.4 gives a simple sufficient conditional-Gaussian specification. No Gaussianity is otherwise imposed.

The class $\mathcal{E}$ separates two forces governing portfolio learnability. The exponent γ controls the persistence of the data, while b controls the geometry of the managed-portfolio spectrum. The parameter b describes how concentrated the managed-portfolio opportunity set is. A large b means that most payoff variation is concentrated in a relatively small number of strong portfolio directions. A smaller b leaves a longer tail of economically distinct but progressively weaker directions.

Polynomial spectral decay is natural for finite-smoothness representations, including Sobolev–Matérn, Laplace, and spline-type kernels. Gaussian kernels can instead generate exponential or faster-than-polynomial decay (Belkin, 2018). For exposition, we use $b = \infty$ as shorthand for the

fast-decay benchmark, including Gaussian-type spectra and the finite-rank limit. The class $\mathcal{E}$ itself has finite b; these are separate benchmarks, not substitutions into the restriction $\gamma > b$. Their relevance depends on the spectrum of $\Sigma_{\mathcal{H}_K,\varepsilon}$, not on the kernel name alone.

B. Learnable Sharpe ratio and optimal shrinkage

We can now ask how regularization should change as market history accumulates, how rapidly the population maximum-Sharpe direction can be recovered, and how much of the managed-portfolio opportunity set can be reliably used. The financial observations can be independent over time or weakly dependent.

Population loss and Sharpe of an estimated policy hold the training sample fixed and integrate over the stationary one-period evaluation law. They are not the Sharpe ratio of a finite test sample.

Theorem 1 (Optimal shrinkage and learnable frontier)**.** *Choose the optimal regularization*

$$\lambda_T^\star \asymp T^{-b/(b+1)}$$

, with comparison constants independent of the economy. Uniformly over $\varepsilon \in \mathcal{E}$,

$$Q_\varepsilon(\widehat{W}_{\lambda_T^\star,T}) - Q_\varepsilon(W_\varepsilon^\star) = O_{\mathbb{P}_\varepsilon}\left(T^{-b/(b+1)}\right). \tag{9}$$

Moreover, uniformly over $\varepsilon \in \mathcal{E}$,

$$\mathrm{SR}_{\varepsilon,K}^\star - \mathrm{SR}\left(R_{t+1}^p(\widehat{W}_{\lambda_T^\star,T})\right) = O_{\mathbb{P}_\varepsilon}\left(T^{-b/(b+1)}\right), \tag{10}$$

where:

$$SR_{\varepsilon}\left(\widehat{W}_{K,\lambda,T}\right) = \frac{\mathbb{E}_{\varepsilon}\left[R^{p}_{t+1}\left(\widehat{W}_{K,\lambda,T}\right)\middle|\widehat{W}_{K,\lambda,T}\right]}{\sqrt{\mathrm{Var}_{\varepsilon}\left(R^{p}_{t+1}\left(\widehat{W}_{K,\lambda,T}\right)\middle|\widehat{W}_{K,\lambda,T}\right)}} = \frac{\mathbb{E}_{\varepsilon}\left[\frac{1}{N}\sum_{i=1}^{N}\widehat{W}_{K,\lambda,T}(Z_{i,t})R_{i,t+1}\middle|\widehat{W}_{K,\lambda,T}\right]}{\sqrt{\mathrm{Var}_{\varepsilon}\left(\frac{1}{N}\sum_{i=1}^{N}\widehat{W}_{K,\lambda,T}(Z_{i,t})R_{i,t+1}\middle|\widehat{W}_{K,\lambda,T}\right)}}$$

and

$$SR^{\star}_{\varepsilon,K} = \sup_{W\in\mathcal{H}_K}\frac{\mathbb{E}_{\varepsilon}\left[\frac{1}{N}\sum_{i=1}^{N}W(Z_{i,t})R_{i,t+1}\right]}{\sqrt{\mathrm{Var}_{\varepsilon}\left(\frac{1}{N}\sum_{i=1}^{N}W(Z_{i,t})R_{i,t+1}\right)}}.$$

The theorem first establishes consistency of direct portfolio learning. Since $\lambda^{\star}_{T} \to 0$ and $\mathrm{SR}^{\star}_{K} - \mathrm{SR}\left(\widehat{W}_{\lambda^{\star}_{T},T}\right) \to_{P} 0$, the portfolio manager eventually recovers the maximum-Sharpe direction available within the chosen portfolio representation.

The optimal regularization path also gives a dynamic interpretation to the eigenportfolio logic discussed above. With a short return history, directions deep in the managed-portfolio spectrum are weakly identified and are therefore shrunk heavily. As more observations become available, their expected payoff and payoff variation can be estimated more precisely, allowing the manager to take progressively larger positions in those directions.

Indeed, as $\lambda^{\star}_{T} \asymp T^{-b/(b+1)}$ falls with T, the spectral filter $\frac{\mu_j}{\mu_j+\lambda^{\star}_{T}}$ moves toward one for progressively smaller values of μ_j. Directions that were effectively suppressed in short samples therefore become statistically usable in longer samples.

Regularization is thus a finite-history device. More data allow the portfolio manager to move deeper into the managed-portfolio spectrum and exploit investment opportunities that were previously too weakly identified.

The shape of the spectrum determines how quickly portfolio opportunities can be learned. A steep spectrum (large b) concentrates economically relevant payoff variation in a few directions, making the portfolio easier to learn. A flatter spectrum spreads variation across many weaker directions, increasing potential breadth but slowing learning.

Importantly, portfolio learnability is therefore not determined directly by the number D of firm characteristics. Two representations with the same number of characteristics can generate very different learning problems. If their characteristics produce largely redundant portfolio bets, the managed-portfolio spectrum is concentrated. If instead they produce many distinct but weak portfolio directions, the spectrum is diffuse and substantially more market history is required.

What matters for the portfolio manager is therefore not the raw number of signals, but the economic geometry of the portfolios generated by those signals.

Figure 1 gives a graphical interpretation of the learning result: as market history grows, the learned capital-allocation line rotates toward the population tangency line. Under Condition (E6), at any fixed volatility, the expected-return gap closes at rate $O_P\left(T^{-b/(b+1)}\right)$.

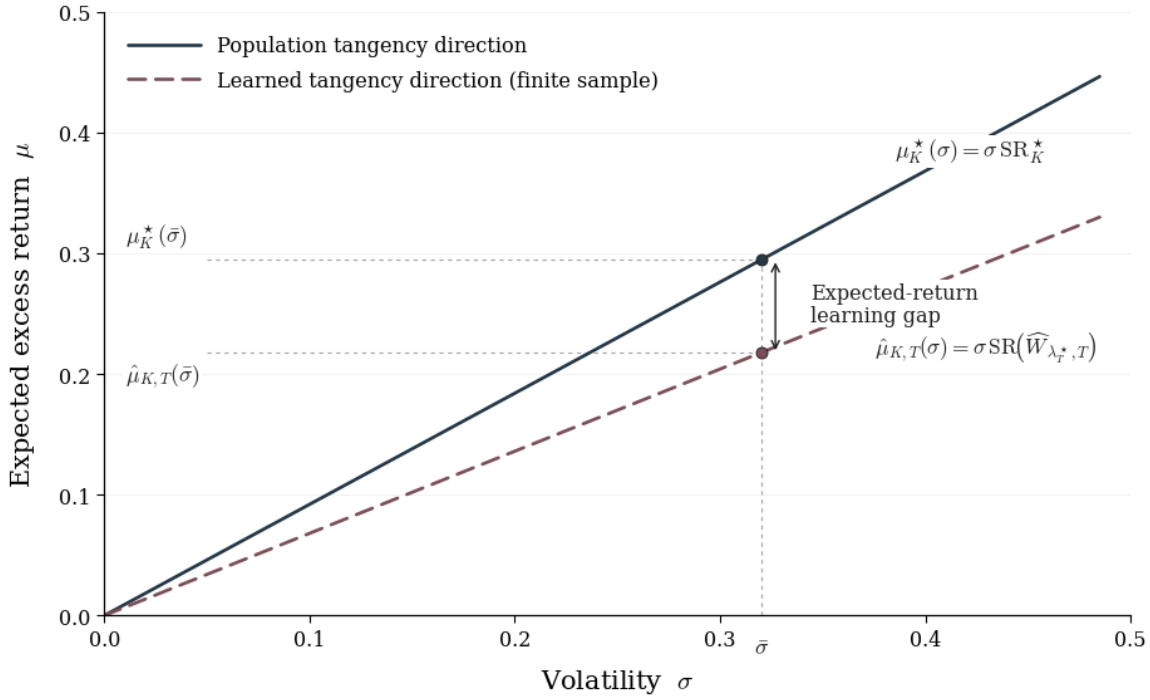


Figure 1. Learning the tangency direction. The population tangency direction generates expected excess return $\mu_K^\star(\sigma) = \sigma\,\mathrm{SR}_K^\star$, while the direction learned from a finite history generates $\hat\mu_{K,T}(\sigma) = \sigma\,\mathrm{SR}(\widehat{W}_{\lambda_T^\star,T})$. At the fixed volatility target $\bar\sigma$, the vertical distance is the expected-return learning gap $\mu_K^\star(\bar\sigma) - \hat\mu_{K,T}(\bar\sigma)$.

C. Optimal effective portfolio complexity

The optimal-shrinkage result can now be expressed directly in terms of portfolio complexity.

Theorem 2 (Learnable effective portfolio complexity)**.** *Under the portfolio-loss conditions of Theorem 1, the rate-optimal effective complexity satisfies*

$$C_{K,T}^\star := C_K(\lambda_T^\star) \asymp T^{1/(b+1)}. \tag{11}$$

The theorem gives a direct economic interpretation of additional market history. As T increases and optimal shrinkage declines, the portfolio manager can reliably activate a broader set of managed-portfolio directions.

This expansion is nevertheless gradual. Effective complexity grows sublinearly: $\frac{C_{K,T}^\star}{T} \to 0$.

Thus, even when the underlying representation is arbitrarily rich, a finite market history supports only a limited fraction of its potential portfolio directions.

A steep managed-portfolio spectrum implies slower growth of effective complexity because most economically relevant payoff variation is already concentrated in a few strong directions. A flatter spectrum supports greater potential breadth, but exploiting that breadth requires substantially more market history.

Figure 2 summarizes this tradeoff. With a short history, the Sharpe-maximizing portfolio uses only a small number of effective directions. With more observations, weaker directions become reliably investable and the optimal effective complexity shifts outward.

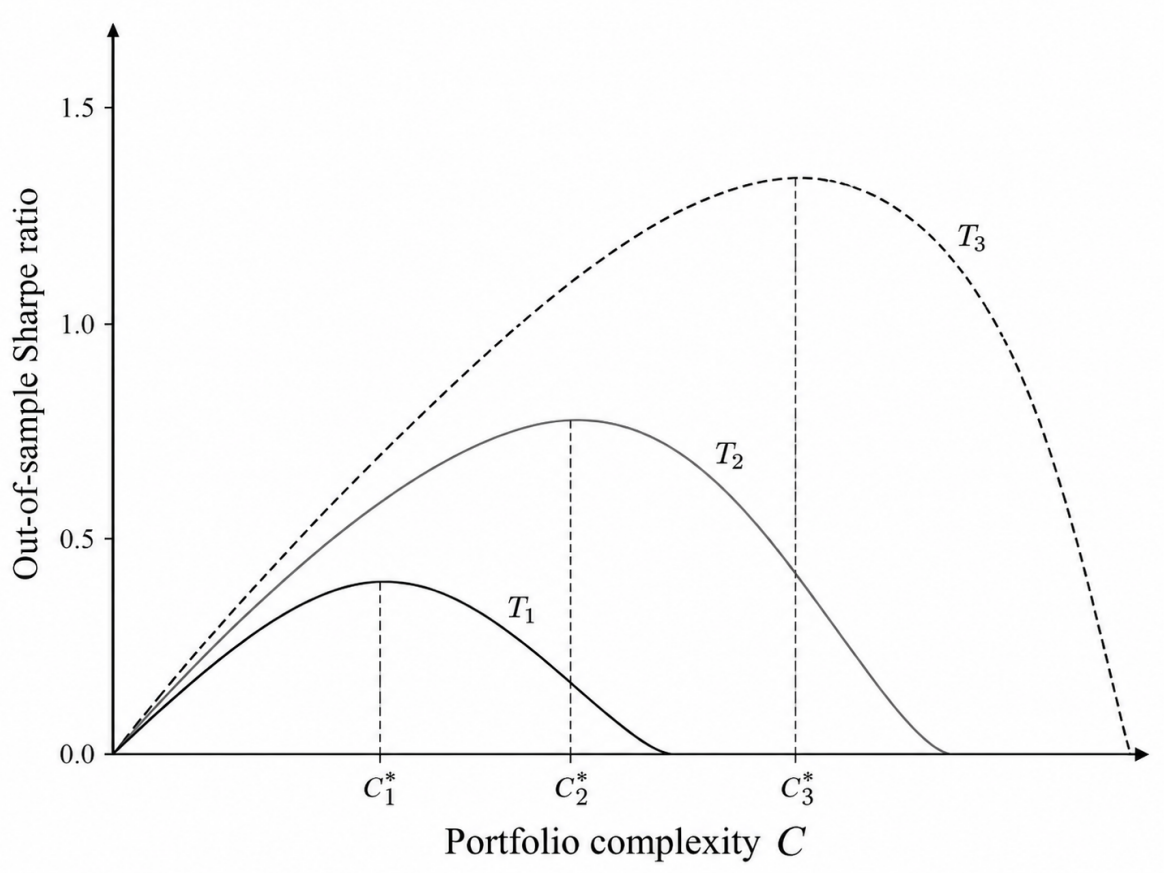


Figure 2. Market history and learnable portfolio complexity. Schematic Sharpe–complexity profiles for different history lengths. Longer market histories allow the portfolio manager to activate a broader set of managed-portfolio directions, shifting the Sharpe-maximizing effective complexity to the right.

The distinction from nominal size is central. A large representation expands the set of portfolio policies that can potentially be expressed. Effective complexity measures how much of that representation the portfolio manager can actually use given the available history of returns. Effective complexity is therefore a learnable economic that quantify the amount of eigenportoflios that the portoflio manager trade.

D. The fundamental limit

The previous results describe how a portfolio manager should use a finite market history: weak managed-portfolio directions should be shrunk, the amount of shrinkage should decline as more observations become available, and the set of portfolio opportunities that can be reliably exploited expands with the sample. A natural question remains. Could a different learning technology overcome this limitation? Could a sufficiently large neural network, a richer random-feature model, or a more sophisticated portfolio algorithm recover the optimal investment decision substantially faster?

The answer is no without additional economic structure.

Theorem 3 (Fundamental limit of maximum-Sharpe learning)**.** *There exists a constant $c > 0$ such that, for all sufficiently large T,*

$$\inf_{\widetilde{W}_T} \sup_{\varepsilon \in \mathcal{E}} \mathbb{E}_\varepsilon \left[\mathrm{SR}^\star_{\varepsilon,K} - \mathrm{SR}_\varepsilon\left(R^p_{t+1}(\widetilde{W}_T) \right) \right] \geq c\, T^{-b/(b+1)}.$$

The infimum is over all measurable portfolio-learning procedures based on the T observed financial rows. Hence no portfolio-learning procedure can uniformly recover the population maximum Sharpe ratio at a faster rate than $T^{-b/(b+1)}$ over $\mathcal{E}$.

Together with Theorem 1, this establishes $T^{-b/(b+1)}$ as the minimax-optimal rate for learning the maximum Sharpe ratio over the class considered here.

The limitation is therefore informational rather than computational. Portfolio directions deep in the managed-payoff spectrum are only weakly identified by a finite history of realized returns. A richer model may represent more investment opportunities, and a better algorithm may use the available information more effectively, but neither can make those weakly identified opportunities more visible in the data.

Learning faster requires knowing more about the economy. This can come from a longer market history or from additional economic restrictions on the return process, the managed-payoff spectrum, or the location of optimal portfolio demand. Absent such information, the learning rate is a

property of the investment problem, not of the particular algorithm used to solve it.

V. Representation, Learnability, and the Virtue of Complexity

Kelly et al. (2024) study an environment in which expanding the model also captures an increasing share of the underlying economic signal. In their setting, the improvement in approximation can dominate the additional estimation cost. This is the virtue of complexity.

Our question is different. We hold the investor's information set fixed and vary only the richness of the portfolio representation. We take the view that a portfolio manager should make available all information she considers economically useful and can observe in real time. Regularization, rather than ex ante feature exclusion, should then determine how much of the resulting portfolio opportunity set can actually be learned from the available data. We therefore study the virtue of Complexity, as the virtue of richer representation, separately from the addition of new signals.

Our point in this section is that a richer portfolio representation and a better learned portfolio are two different objects.

Let $\mathrm{SR}^\star$ denote the maximum population Sharpe ratio over the space of L^2 portfolio policies,

$$\mathrm{SR}^\star := \sup_{W \in L^2} \mathrm{SR}(W),$$

and let

$$\mathrm{SR}_K^\star := \sup_{W \in \mathcal{H}_K} \mathrm{SR}(W)$$

denote the maximum population Sharpe ratio available within representation K, where $\mathcal{H}_K \subseteq L^2$ is the corresponding policy space. For example, a linear kernel restricts portfolio weights to be linear functions of characteristics, whereas a Matérn kernel allows a richer nonlinear class.

Finally,

$$\mathrm{SR}(\widehat{W}_{K,\lambda,T})$$

denotes the population Sharpe ratio of the portfolio policy learned within representation K from a market history of length T.

The gap between the unrestricted population opportunity set and the portfolio actually learned from the data can be decomposed as

$$\mathrm{SR}^\star - \mathrm{SR}(\widehat{W}_{K,\lambda,T}) = \underbrace{\mathrm{SR}^\star - \mathrm{SR}^\star_K}_{\text{representation gap}} + \underbrace{\mathrm{SR}^\star_K - \mathrm{SR}(\widehat{W}_{K,\lambda,T})}_{\text{learning gap}}. \tag{12}$$

The representation gap measures the investment opportunities that cannot be expressed by the chosen representation, even if the investment environment were known perfectly. The learning gap measures how much of the opportunity set available within that representation cannot be recovered from a finite history of returns.

Now consider two nested representations,

$$\mathcal{H}_{K_1} \subseteq \mathcal{H}_{K_2}.$$

At the population level,

$$\mathrm{SR}^\star_{K_2} \geq \mathrm{SR}^\star_{K_1}.$$

A richer representation can therefore only expand the population investment opportunity set. In this precise sense, complexity is a virtue for representation.

But this population ordering need not survive estimation from finite data. It is perfectly possible that

$$\mathrm{SR}^\star_{K_2} > \mathrm{SR}^\star_{K_1}$$

while

$$\mathrm{SR}(\widehat{W}_{K_2,\lambda_2,T}) < \mathrm{SR}(\widehat{W}_{K_1,\lambda_1,T}).$$

The larger representation contains better population opportunities, but some of those opportunities may lie in managed-portfolio directions that are too weakly identified by the available return history.

A richer representation therefore expands what the portfolio manager could exploit. Whether

those additional opportunities improve the learned portfolio depends on whether they can actually be learned from the available data.

VI. Simulations: a constant calibrated once

We use the frozen Rich6D economic experiment: a six-dimensional Matérn-3/2 representation, Fourier truncation $M = 128$, and 300 independent histories in each of five paired economies. The baseline has $N = 600$ and $\rho = 0.95$; alternatives change N to 300 or 1200, or persistence to 0 or 0.75. Analytic economic optima are specific to N. Sharpe ratios are marginal monthly population quantities multiplied by $\sqrt{12}$; response-one loss is not annualized.

For each economy and history, initial cross-validation selects one constant from the unchanged 71-point grid $[10^{-8}, 10^{-1}]$ using observations 1–60 only. Four forward folds train on 20, 30, 40, and 50 observations and validate on the subsequent ten. Each fold uses $\lambda = an_{\text{train}}^{-0.6}$ and observation-weighted response-one loss; exact ties favor larger a. The selected constant is then frozen:

$$\lambda_{k,T} = \widehat{a}_{k,60} T^{-0.6}.$$

Later observations refit the policy, with no further calibration. The historical constant is a comparison. Oracle Q minimizes mean regret and oracle SR maximizes mean Sharpe on the original 96-penalty grid, choosing one common penalty per economy and horizon. Both use population information and are diagnostics only.

Table I. Baseline endpoint at $T = 4860$, all 300 histories.

Specification	Annual SR	Regret	Empirical C
Initial CV	2.93250	0.01130	143.78233
Historical fixed	2.93477	0.02220	14.95632
Oracle Q	2.95181	0.00833	58.32002
Oracle SR	2.95343	0.00866	41.74288

Initial CV reduces baseline mean endpoint regret from 0.022205 to 0.011296 (paired change -0.010909, 95% Monte Carlo mean interval $[-0.011399, -0.010419]$). Annualized Sharpe changes from 2.934774 to 2.932497 (paired change -0.002277, interval $[-0.005079, 0.000525]$). Empirical complexity rises from 14.956 to 143.782. Thus the response-one improvement does not es-

tablish a final Sharpe improvement. Oracle Q and oracle SR yield distinct endpoint penalties and outcomes; they do not tune the strategy.

The financial components uniquely identified from each policy's signed Sharpe and regret explain this mean discrepancy. At $T = 4860$, mean positive-ray loss is 0.011119 under initial CV versus 0.010733 historically, while mean scale loss falls from 0.011472 to 0.000177. These add to the respective mean regrets. Thus improved scaling more than offsets worse directional fit in the response-one objective. These quantities require no choice of a payoff-moment root; actual monthly moments and the remaining radial decomposition are recovered separately on the audit subset.

The full-grid descriptive slopes of log mean regret and log mean Sharpe gap are -0.437820 and -0.398648, respectively. The unchanged windows $T \geq 360$, $T \geq 720$, and $T \geq 1440$, full cross-horizon covariance MCSE, jackknife checks, and normalized-outcome quantiles are reported in the appendix. No horizon or replication was removed to approach -0.6. The asymptotic statement is an $O_{\mathbb{P}}$ upper bound, not an equality for a finite Monte Carlo mean.

The ideal infinite Fourier target, the simulated $M = 128$ economy, and the fixed-dimensional Nyström estimator are distinct objects. The attached derivation supports maximal source order $r = 1$ for the ideal target and preserves the kernel upper rate under a frozen selection on a finite positive grid, conditional on the manuscript assumptions. It does not certify the numerical approximation. A fixed subspace with positive representation error cannot have regret to the full economic optimum converging to zero. At the baseline endpoint the rank-4096 representation floor is about 15.7% of mean CV regret, exceeding the unchanged 5% criterion. The paired CV rank and quadrature audit remains unresolved; its scope and failed cells are tabulated in the appendix.

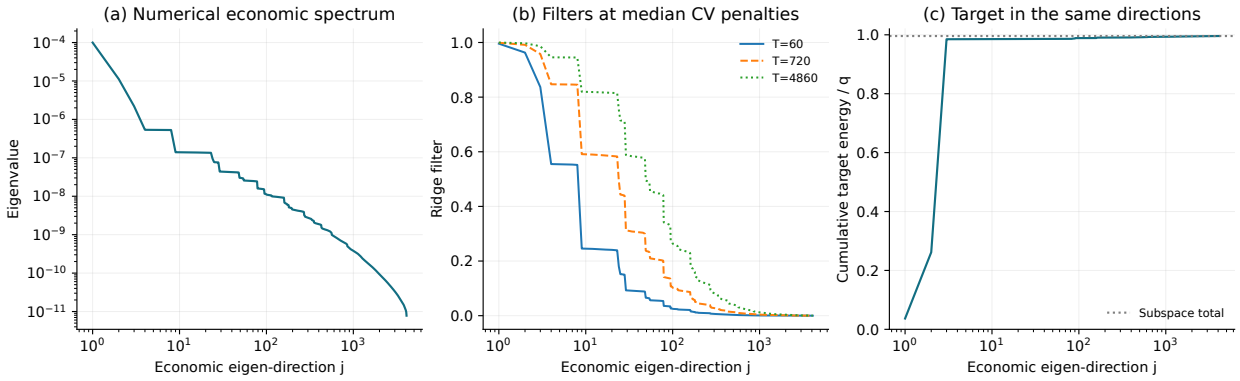


Figure 3. Economic spectrum, ridge filters at median initial-CV penalties, and target energy measured in the same numerical eigen-directions. The displayed operator is the frozen rank-4096 approximation; it is not an infinite-dimensional spectrum.

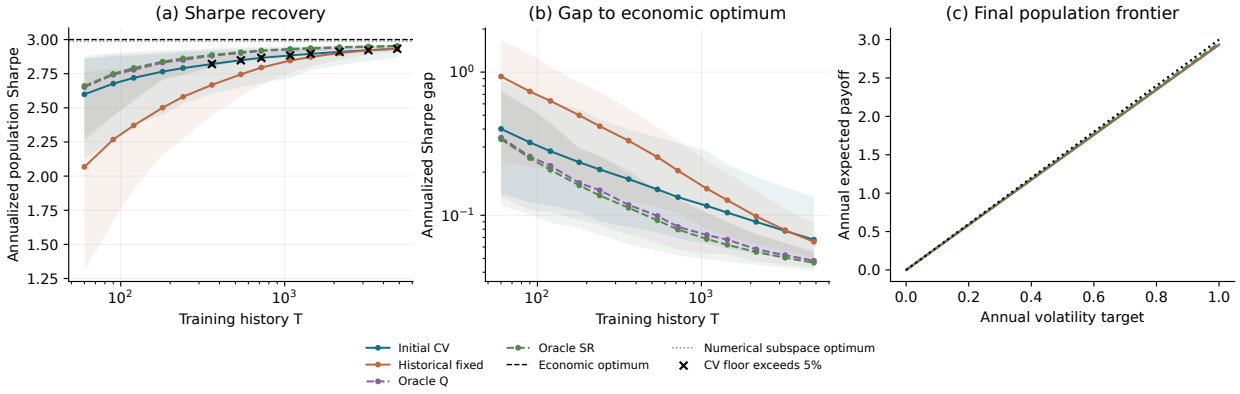


Figure 4. Baseline population Sharpe recovery, gap to the analytic economic optimum, and final capital-allocation frontiers. Curves include every archived horizon. Shaded bands are the central 95 percent of replication outcomes, not confidence intervals of the mean; oracle bands are descriptive and not selection adjusted. Crosses mark CV projection floor/mean regret exceeding 5 percent. The CV audit remains scope-limited; rank and quadrature uncertainty are additional.

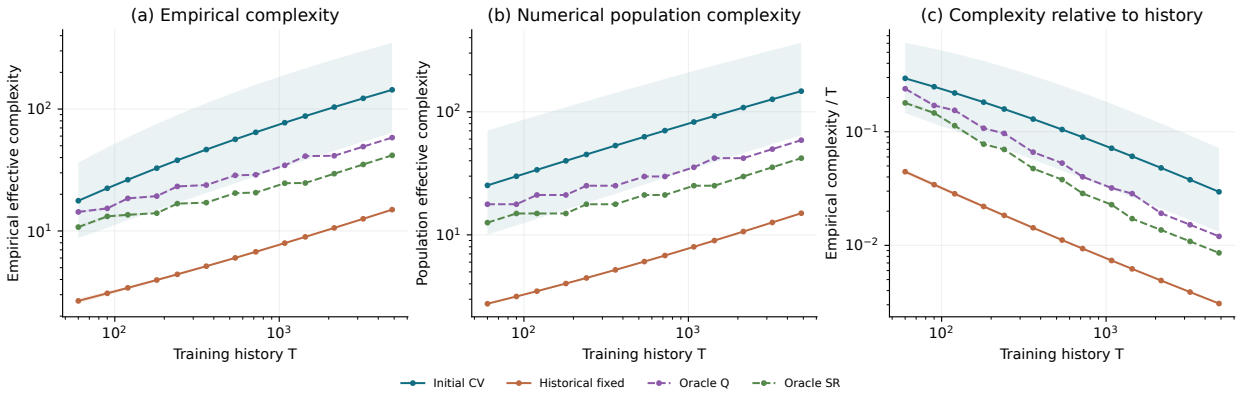


Figure 5. Empirical and numerical population effective complexity, and empirical complexity divided by history length. Population complexity is evaluated at each actual selected penalty before aggregation; it is not complexity at the mean penalty. Bands are replication percentiles.

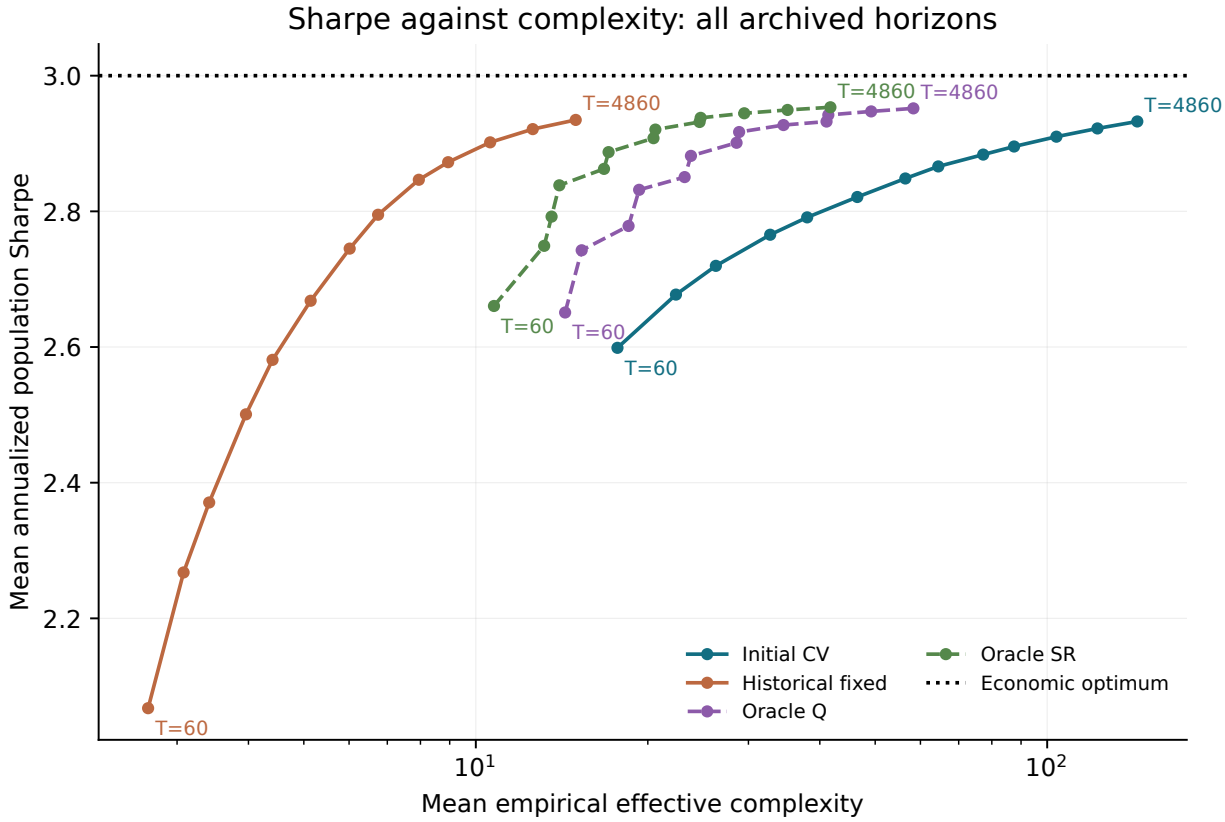


Figure 6. Mean population Sharpe against mean empirical complexity across the full baseline horizon grid. Solid paths implement frozen initial CV or the historical rule; dashed oracle paths use population information. The horizontal line is the true economic optimum.

VII. Empirical analysis

We use the U.S. stock-level data of Jensen, Kelly, and Pedersen (2023). Inspired by Section 2.5 of Didisheim, Ke, Kelly, and Malamud (2024), we retain 130 of the 153 characteristics by coverage using only the initial training sample, 1963–1972. Validation and test observations cannot affect selection. The published candidate dictionary is a fixed retrospective research specification; we do not claim a historical data vintage or contemporaneous signal discovery. We retain common stocks on the main U.S. exchanges, excluding nano stocks, apply the 30% row-missingness threshold, and exclude stock-months with missing JKP next-month excess returns before monthly ranking and the calculation of N_t. This complete-case convention is our sample choice, not a rule attributed to Didisheim et al.; the sample is conditional on future payoff availability. We rank observed characteristics into $[-0.5, 0.5]$ and assign residual missing characteristics neutral zero. We do not impute missing payoffs.

Training expands from 1963, with the preceding five formation years reserved for validation and annual refits at the January formation close. The experiment contains 47 test years and 564 formation months, January 1978–December 2024; realized payoffs run from February 1978 to January 2025. Ridge penalties are selected using the portfolio criterion introduced earlier, along 120-point grids fixed using only the initial 1963–1972 managed-payoff spectra.

Linear is the finite-dimensional benchmark. Gaussian represents smooth nonlinear policies; Matérn-3/2 permits a broader nonlinear class. Each nonlinear policy uses one fixed map of 10,000 random Fourier features, seed zero. Both share a lengthscale of 4.3921, the median distance among 1,000 initial-training vectors. A common fixed positive scale targets median gross exposure 1.8 for the initial Linear validation policy. The last validation payoff becomes known at the January 1978 formation close, before the first OOS payoff.

Figure 7 and Table II summarize the new run. Linear: Sharpe 3.09, maximum drawdown -15.8%; Gaussian: Sharpe 3.30, maximum drawdown -15.6%; Matérn-3/2: Sharpe 3.46, maximum drawdown -13.6%. Total wealth compounds the cash return plus the scaled portfolio excess payoff,

adding cash exactly once. Drawdown uses exactly that wealth path, including its starting value.

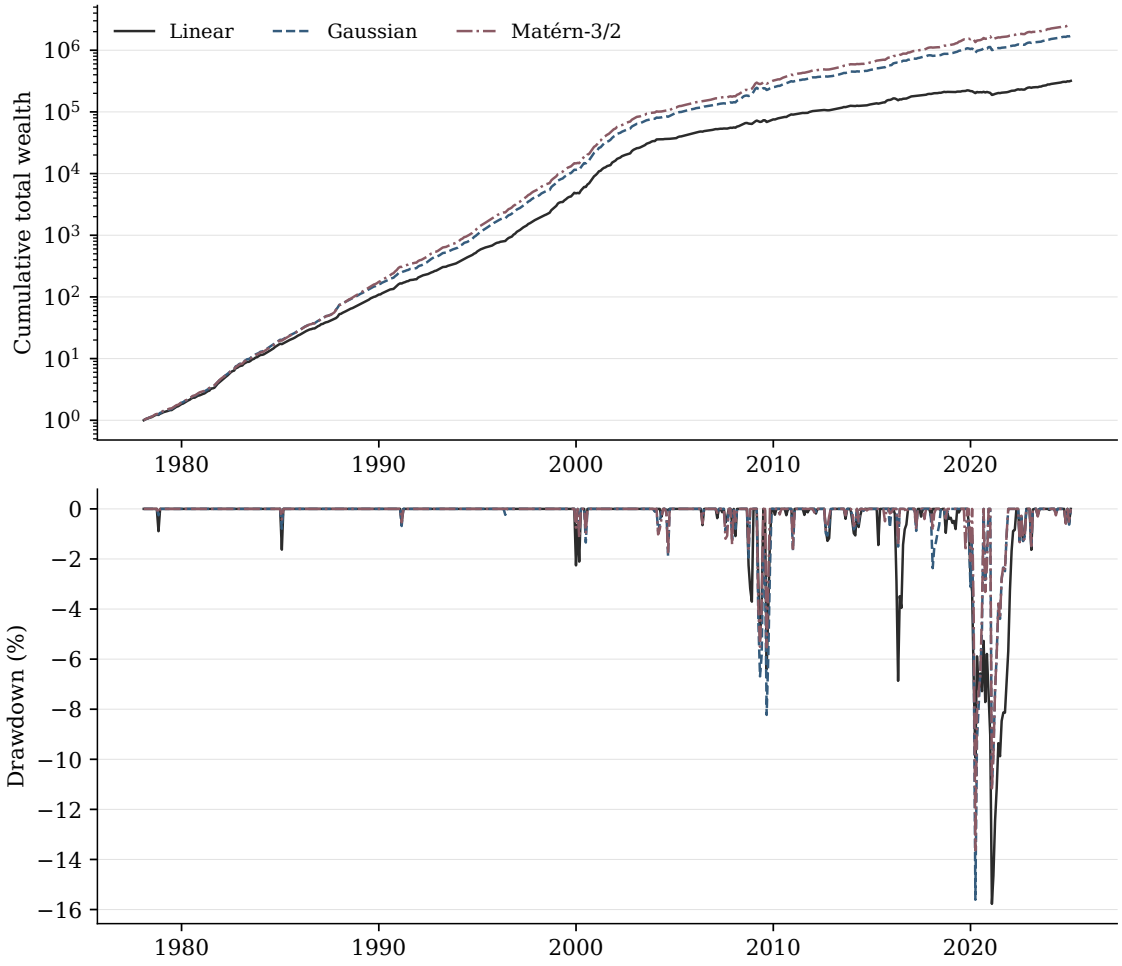


Figure 7. Cumulative total wealth (log scale) and drawdown from the same wealth series. All policies use the same fixed pre-payoff scale.

Figure 8 reports gross and net stock exposure under the same scale.

Representation determines which investment opportunities can potentially be expressed. The managed-portfolio spectrum describes their payoff structure. Effective complexity determines how much of that spectrum is actually used after regularization. Figure 9 shows the ordered positive second-moment eigenvalues on the common history preceding the 2024 test window.

Figures 10 and 11 plot historical and subsequent 2024 test Sharpe against $C(\lambda) = \sum_j \mu_j/(\mu_j+\lambda)$

Policy	Annual mean excess	Annual vol.	Sharpe	Max. drawdown	Mean gross	Mean net
Linear	0.234	0.076	3.091	-0.158	2.009	0.132
Gaussian	0.272	0.082	3.300	-0.156	2.326	0.100
Matérn-3/2	0.280	0.081	3.461	-0.136	2.374	0.090

Table II. Performance from the 564 monthly OOS observations. Returns and volatility are annualized from decimal monthly excess returns; maximum drawdown is signed.

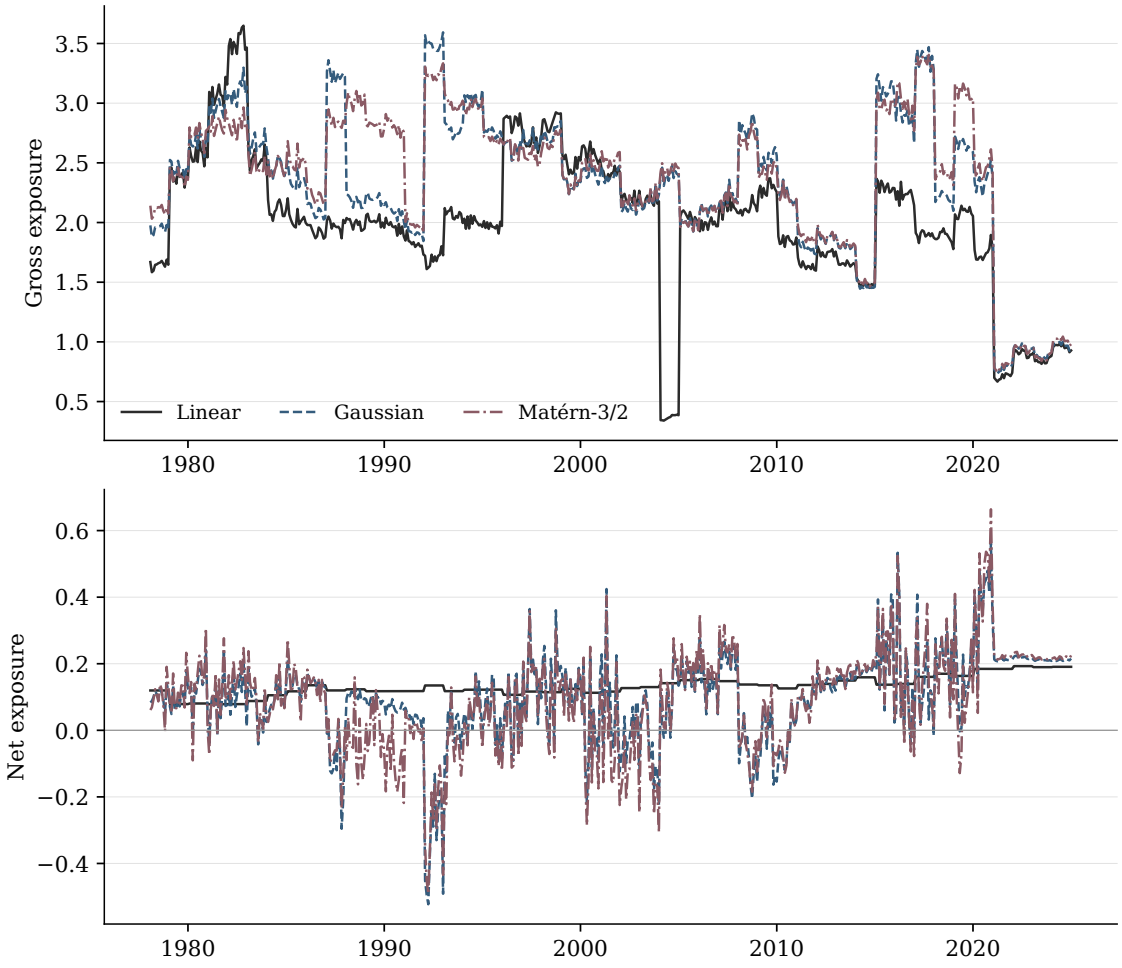


Figure 8. Gross and net exposures for the three policies.

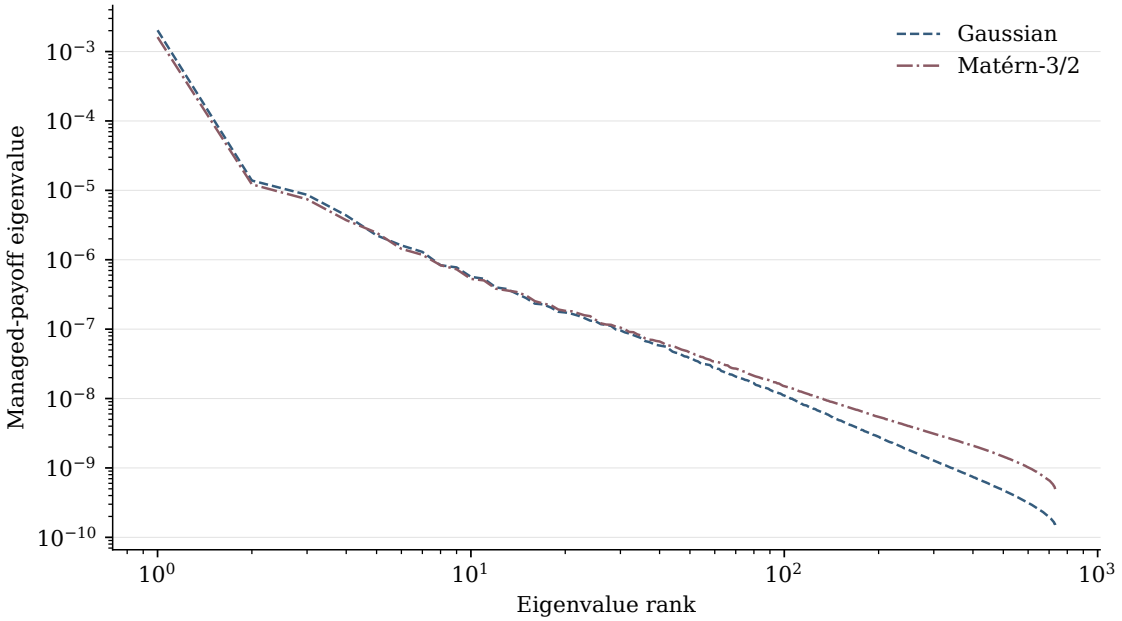


Figure 9. Managed-portfolio spectra on log-log axes.

along the same frozen penalty paths. The test panels use only the twelve payoffs from February 2024 through January 2025. Validation selections are distinguished from descriptive ex-post maxima, which never determine the policy.

An initially calibrated regularization schedule. For the Matérn-3/2 representation, we choose C_0 once using the 1963–1972 training and 1973–1977 formation-validation samples, and use

$$\lambda_T = C_0 T^{-\widehat{b}_T/(\widehat{b}_T+1)}.$$

Here T counts historical monthly managed payoffs, and $\widehat{b}_T$ is re-estimated at each January formation close together with the portfolio coefficients. It is the negative OLS slope of log second-moment eigenvalues on log rank, using the fixed 10%–60% rank band after a relative 10^{-12} numerical cutoff. This finite-sample estimation convention supplements the manuscript's asymptotic rate; it does not establish its population spectral assumption. The main-text rate corresponds to $r = 1$ in the general

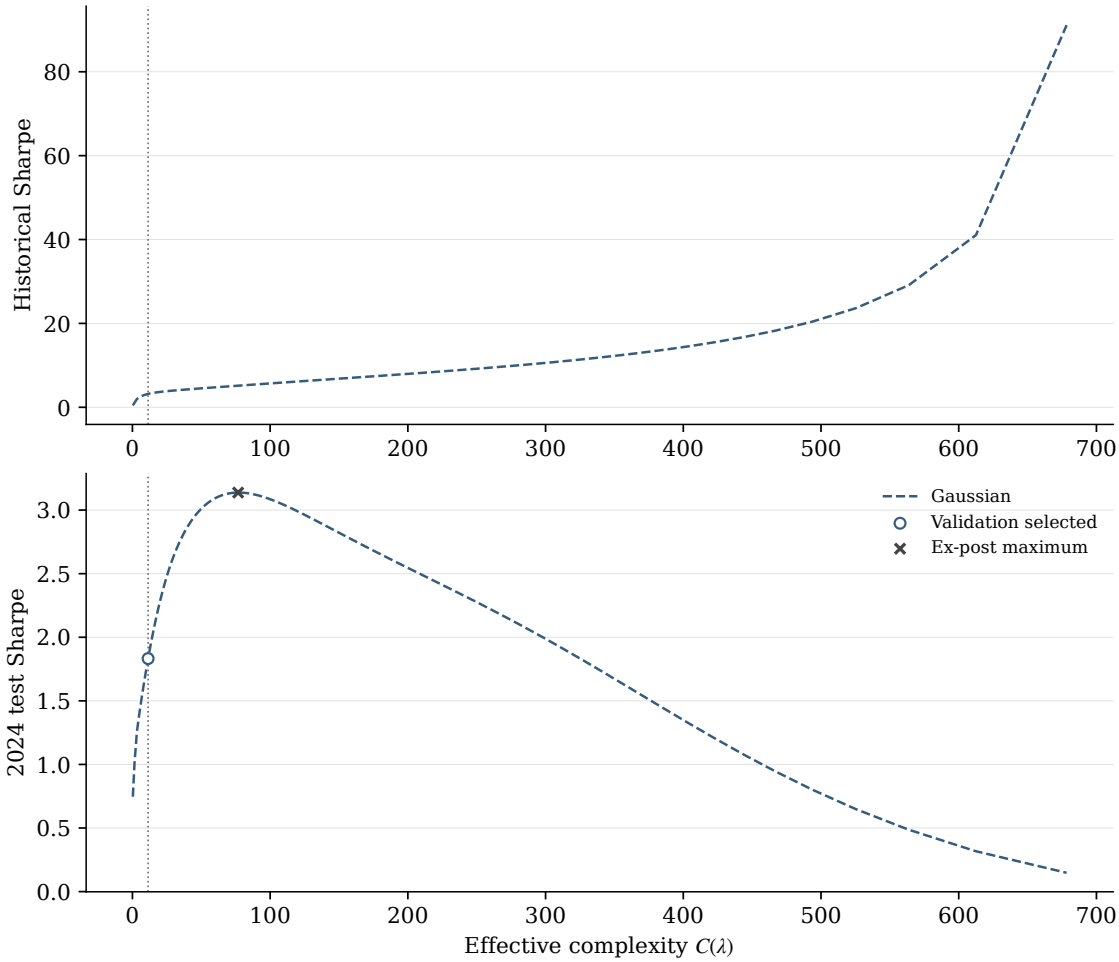


Figure 10. Gaussian effective complexity: historical and true 2024 test Sharpe.

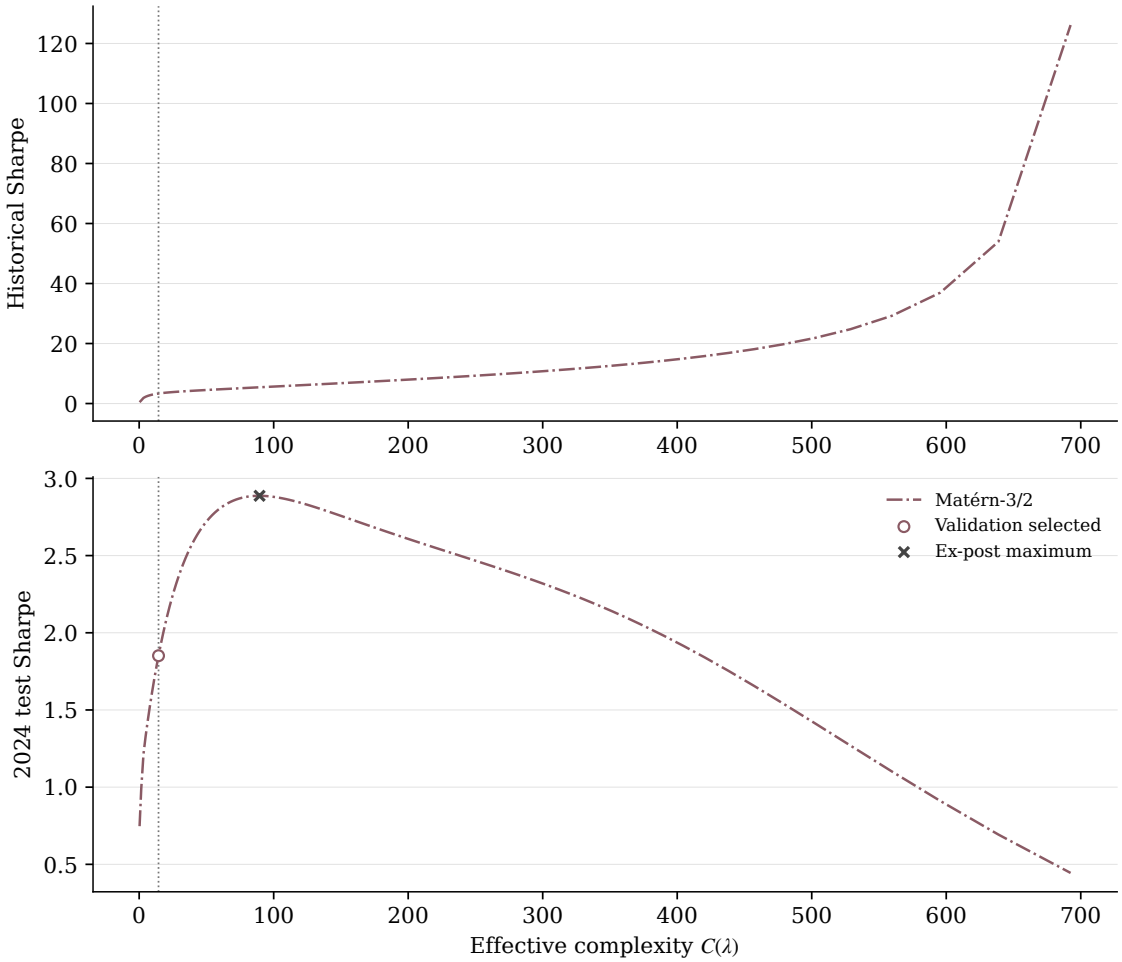


Figure 11. Matérn-3/2 effective complexity: historical and true 2024 test Sharpe.

appendix. The grid retains all 120 previous initial-training penalties and adds 80 intermediate effective-complexity points before converting penalties to C_0. Validation selects $C_0 = 2.265779 \times 10^{-6}$, available on January 31, 1978. The constant remains unchanged during all 47 subsequent annual refits.

Selected effective complexity rises from 36.681 in 1978 to 91.776 in 2024; regularization falls from 1.15864×10^{-7} to 4.67964×10^{-8}. Annual spectral estimates range from 1.3309 to 1.4323. The 564 selected OOS monthly payoffs have an annualized Sharpe of 3.561. This remains an exceptionally strong gross backtest, subject to the same complete-case payoff selection, retrospective characteristic dictionary, current-snapshot and execution assumptions as the main experiment; trading and borrowing costs are not subtracted.

Each annual curve evaluates only its twelve subsequent payoffs, from February through the following January. Annual Sharpe estimates are therefore noisy. Loss is the unscaled held-out mean of $(1 - R^p)^2$, with no test-period ridge penalty and no wealth-plot scaling. Neither the OOS loss nor OOS Sharpe selects C_0, the spectral rank band, or the annual coefficients. Changes in estimated b could make the plug-in regularization path locally nonmonotonic, although no annual increases occur in this sample. No curve shape is imposed.

The displayed paths use 1,202 diagnostic values per year over an interval extended by a factor of 1,000 at each end, while the initial policy selection continues to use the frozen 200 candidates. A validation-only search over 3,001 points and numerical refinement improves the selected validation loss by only 0.0004464%. Additional evaluations at all 56,447 diagnostic midpoints bound the largest observed interpolation discrepancy at 0.00112 Sharpe points and 0.000057 loss units. The exact $\lambda = 0$ minimum-norm solution is included and marked on every curve; its empirical complexity is the historical rank, equal to T in all 47 windows. Its initial-validation loss is 0.820087, compared with 0.598552 at the selected positive penalty. Both weak- and strong-regularization limits are checked separately. Extrema at a boundary are not claimed to be interior optima. Companion figures divide complexity by each fit's historical monthly sample size T: the selected ratio falls from 0.2038 to 0.1254.

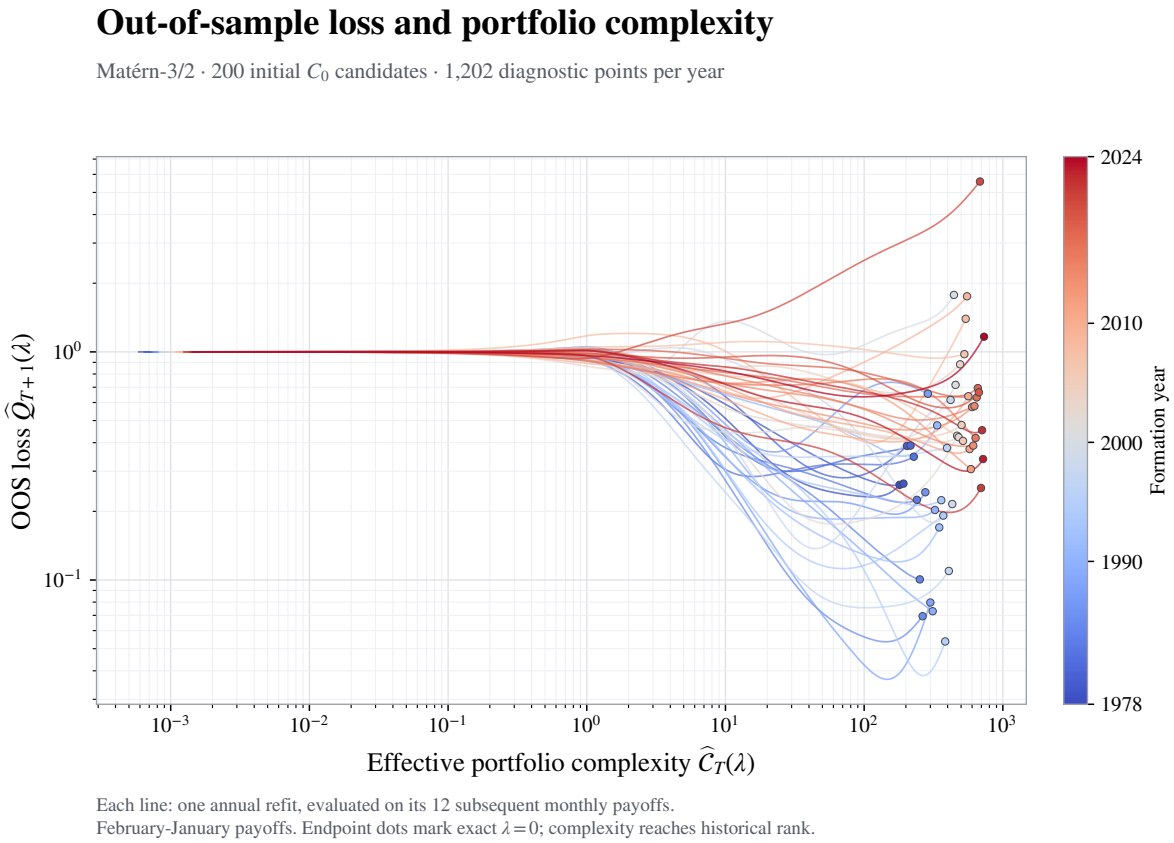


Figure 12. Next-year OOS quadratic loss against effective complexity for every annual Matérn refit, 1978–2024. Colors indicate formation years.

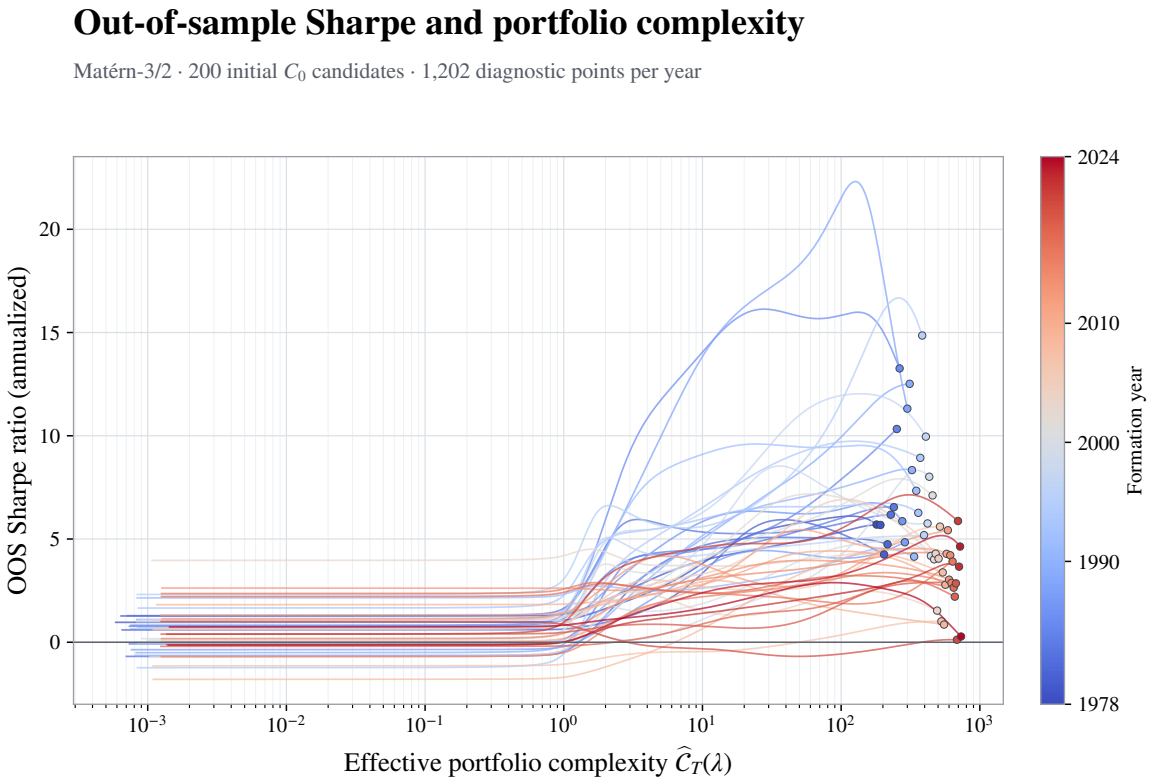


Figure 13. Annualized Sharpe against effective complexity, computed separately from each twelve-month OOS window.

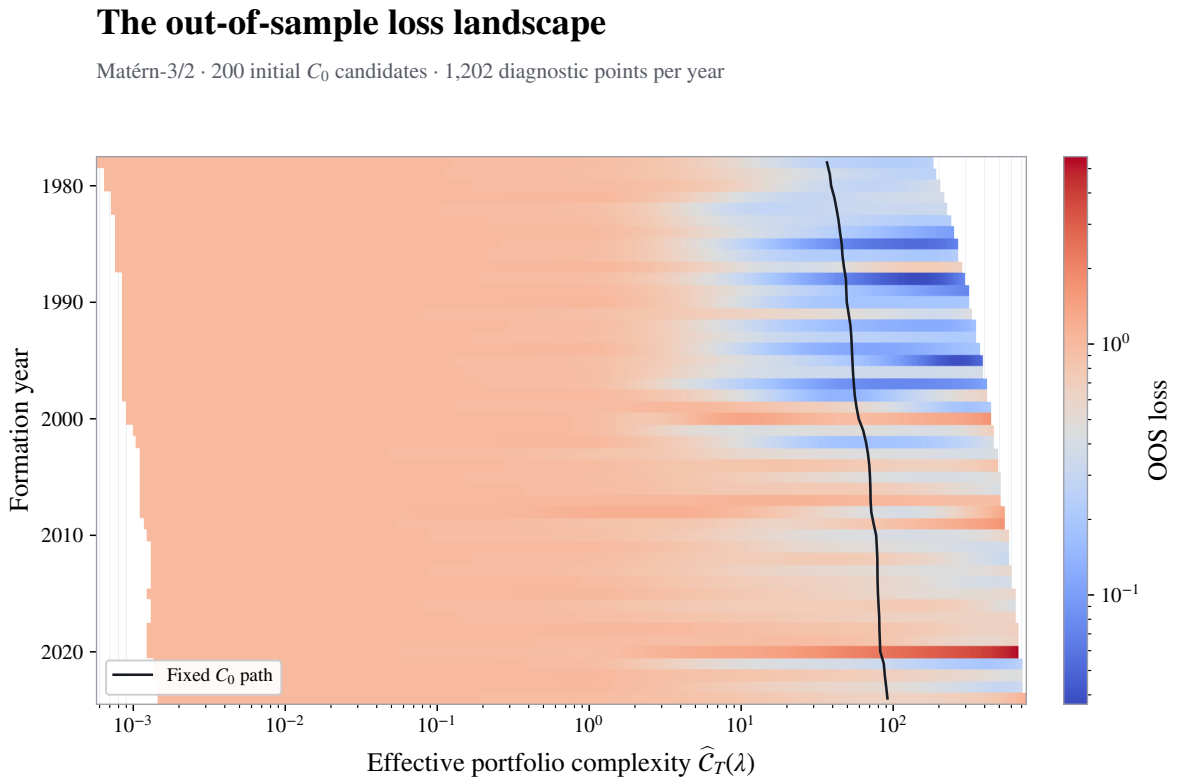


Figure 14. OOS loss by formation year and effective complexity. Interpolation is confined to each year's observed complexity support. The black line is the path implied by the initially selected, fixed C_0.

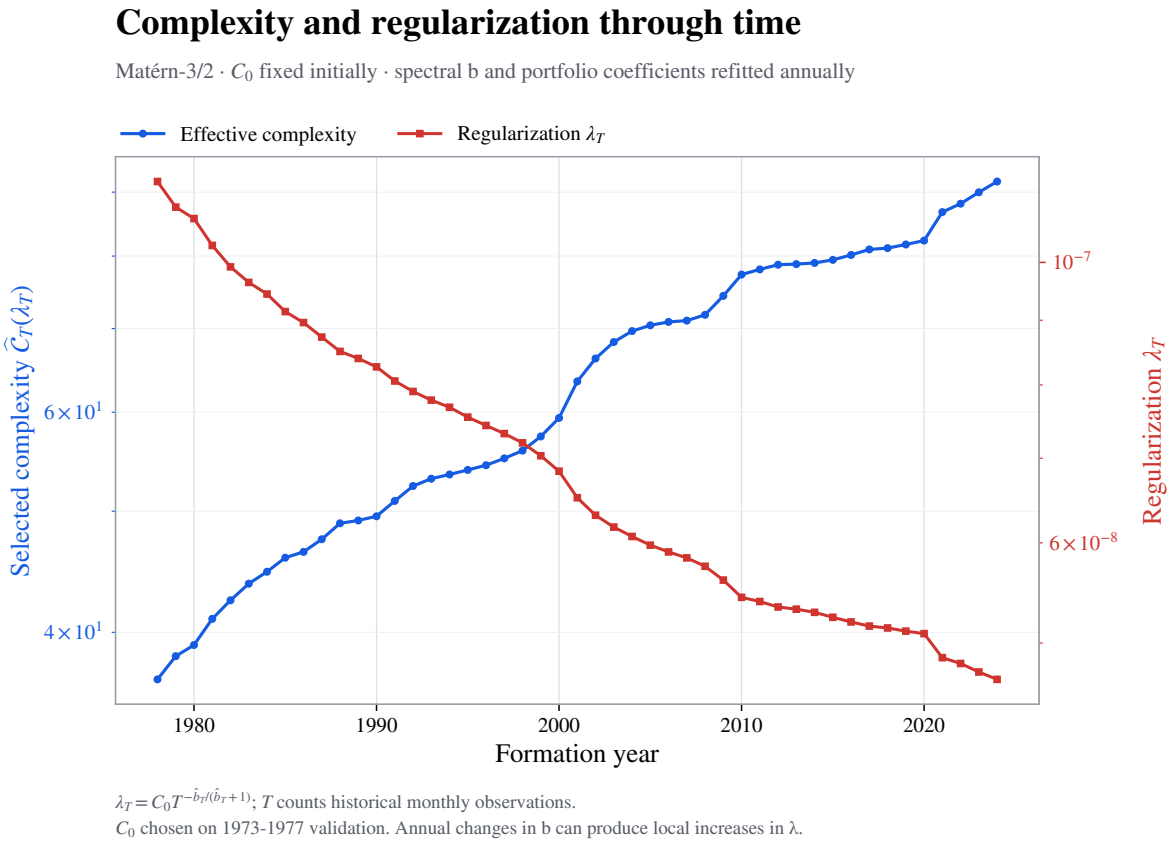


Figure 15. Effective complexity and regularization implied by a fixed initial C_0 and annually updated historical spectral estimates.

References

Belkin, M. (2018). Approximation beats concentration? an approximation view on inference with smooth radial kernels. In *Proceedings of the 31st Conference on Learning Theory*, volume 75 of *Proceedings of Machine Learning Research*, pages 1348–1361.

Bignozzi, E., Gagliardini, P., and Schneider, P. (2026). Optimal rates for kernel ridge regression under β-mixing. Working paper.

Brandt, M. W. (1999). Estimating portfolio and consumption choice: A conditional euler equations approach. *The Journal of Finance*, 54(5):1609–1645.

Brandt, M. W., Santa-Clara, P., and Valkanov, R. (2009). Parametric portfolio policies: Exploiting characteristics in the cross-section of equity returns. *The Review of Financial Studies*, 22(9):3411–3447.

Britten-Jones, M. (1999). The sampling error in estimates of mean–variance efficient portfolio weights. *The Journal of Finance*, 54(2):655–671.

Caldeira, J. a. F., Santos, A. A. P., and Torrent, H. S. (2023). Semiparametric portfolios: Improving portfolio performance by exploiting non-linearities in firm characteristics. *Economic Modelling*, 122:106239.

Filipović, D. and Schneider, P. (2024). Joint estimation of conditional mean and covariance for unbalanced panels. Swiss Finance Institute Research Paper Series 24-60, Swiss Finance Institute. Revised March 2025.

Hansen, L. P. and Jagannathan, R. (1991). Implications of security market data for models of dynamic economies. *Journal of Political Economy*, 99(2):225–262.

Jacot, A., Gabriel, F., and Hongler, C. (2018). Neural tangent kernel: Convergence and generalization in neural networks. In *Advances in Neural Information Processing Systems 31*, pages 8571–8580.

Kelly, B. T. and Malamud, S. (2025). Understanding the virtue of complexity. Research Paper 25-96, Swiss Finance Institute. Working paper.

Kelly, B. T., Malamud, S., and Zhou, K. (2024). The virtue of complexity in return prediction. *The Journal of Finance*, 79(1):459–503.

Kozak, S., Nagel, S., and Santosh, S. (2020). Shrinking the cross-section. *Journal of Financial Economics*, 135(2):271–292.

Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1):77–91.

Minsker, S. (2017). On some extensions of Bernstein's inequality for self-adjoint operators. *Statistics & Probability Letters*, 127:111–119.

Nagel, S. (2025). Seemingly virtuous complexity in return prediction. Working Paper 34104, National Bureau of Economic Research.

Rahimi, A. and Recht, B. (2007). Random features for large-scale kernel machines. In *Advances in Neural Information Processing Systems 20*, pages 1177–1184.

Simon, F., Weibels, S., and Zimmermann, T. (2026). Deep parametric portfolio policies. *Management Science*. Articles in Advance.

A. Proofs and Auxiliary Results

A.1. Maximum-Sharpe direction

Proof of Proposition 1. Step 1: separate direction and scale. Every nonzero $W \in \mathcal{H}$ can be written as $W = aU$, with $\|U\|_{\mathcal{H}} = 1$ and $a \in \mathbb{R}$. Conversely, every such aU belongs to $\mathcal{H}$, and $a = 0$ includes the zero policy. Thus this parametrization does not change the feasible set. Since the payoff is linear in the policy,

$$\min_{W \in \mathcal{H}} Q(W) = \min_{\substack{U \in \mathcal{H} \\ \|U\|_{\mathcal{H}}=1}} \min_{a \in \mathbb{R}} \mathbb{E}\left[\left(1 - aR^p_{t+1}(U)\right)^2\right].$$

A zero-payoff direction has loss one for every scale. A nonzero constant payoff would give minimum loss zero after rescaling. The assumption $0 < Q(W^\star) < 1$ therefore allows zero-variance directions to be omitted.

Step 2: optimize the scale. Fix a direction U with positive payoff variance and write

$$m_U := \mathbb{E}[R^p_{t+1}(U)], \qquad s^2_U := \mathbb{E}[(R^p_{t+1}(U))^2] > 0.$$

Expanding the square and differentiating gives

$$Q(aU) = 1 - 2am_U + a^2 s^2_U, \qquad \frac{\partial Q(aU)}{\partial a} = -2m_U + 2as^2_U.$$

The second derivative is $2s^2_U > 0$, so the unique minimizing scale is $a^\star(U) = m_U/s^2_U$. Substitution

yields

$$\begin{aligned}\min_{a\in\mathbb{R}} Q(aU) &= 1 - 2\frac{m_U}{s_U^2}m_U + \left(\frac{m_U}{s_U^2}\right)^2 s_U^2 \\ &= 1 - \frac{m_U^2}{s_U^2} \\ &= \frac{\mathrm{Var}(R_{t+1}^p(U))}{\mathrm{Var}(R_{t+1}^p(U)) + m_U^2} \\ &= \frac{1}{1+\mathrm{SR}^2(R_{t+1}^p(U))}.\end{aligned}$$

Step 3: compare directions. Consequently,

$$\min_{W\in\mathcal{H}} Q(W) = \min_{\substack{U\in\mathcal{H}\\ \|U\|_{\mathcal{H}}=1}} \frac{1}{1+\mathrm{SR}^2(R_{t+1}^p(U))}.$$

The right-hand side is minimized by a maximum-squared-Sharpe direction, because $x \mapsto (1+x)^{-1}$ is strictly decreasing on $[0,\infty)$. The corresponding optimally scaled payoff has mean $m_U^2/s_U^2 > 0$, since the minimum is strictly below one. It therefore attains the positive maximum Sharpe ratio, not its negative orientation. □

A.2. Pricing-kernel interpretation and Hansen–Jagannathan equality

Proof of Proposition 2. Fix an economy and suppress its subscript in this proof.

Step 1: portfolio orthogonality. For any $V \in \mathcal{H}$, expansion of the quadratic criterion gives

$$\begin{aligned}Q(W^\star + aV) = Q(W^\star) &- 2a\mathbb{E}\big[(1 - R_{t+1}^p(W^\star))R_{t+1}^p(V)\big] \\ &+ a^2\mathbb{E}[(R_{t+1}^p(V))^2].\end{aligned}$$

Optimality at $a = 0$ implies $\mathbb{E}[m_{t+1}^\star R_{t+1}^p(V)] = 0$ for every V. In particular, choosing $V = W^\star$ gives

$$\mathbb{E}[R_{t+1}^p(W^\star)] = \mathbb{E}[(R_{t+1}^p(W^\star))^2].$$

Since $m^\star = 1 - R^p(W^\star)$, it follows that

$$\mathbb{E}[m^\star] = \mathbb{E}[(m^\star)^2] = Q(W^\star) =: q^\star \in (0, 1).$$

Proposition 1 then gives $q^\star = 1/(1 + (\mathrm{SR}^\star)^2)$.

Step 2: normalize the pricing kernel. For $\widetilde{m} = m^\star/q^\star$,

$$\mathbb{E}[\widetilde{m}] = 1, \qquad \mathbb{E}[\widetilde{m}^2] = \frac{1}{q^\star}, \qquad \mathrm{Var}(\widetilde{m}) = \frac{1 - q^\star}{q^\star} = (\mathrm{SR}^\star)^2.$$

Step 3: verify the minimum-variance property. Let d be any square-integrable, possibly signed, random variable with $\mathbb{E}[d] = 1$ and $\mathbb{E}[dR^p(V)] = 0$ for every $V \in \mathcal{H}$. Then

$$\mathbb{E}[dm^\star] = \mathbb{E}[d] - \mathbb{E}[dR^p(W^\star)] = 1.$$

Hence $\mathbb{E}[d\widetilde{m}] = 1/q^\star = \mathbb{E}[\widetilde{m}^2]$, and therefore

$$\mathrm{Var}(d) = \mathrm{Var}(\widetilde{m}) + \mathbb{E}[(d - \widetilde{m})^2] \geq (\mathrm{SR}^\star)^2.$$

Thus $\widetilde{m}$ attains the Hansen–Jagannathan bound on the specified managed excess-return space, with no positivity restriction. None of these steps asserts the stronger conditional restriction in Condition (E4). □

A.3. Managed-portfolio Markowitz representation

Proof of Proposition 3. Under the integrability conditions verified in Lemma 1, the reproducing property gives $R^p_{t+1}(W) = \langle W, X_t \rangle_{\mathcal{H}_K}$. Therefore

$$Q(W) = 1 - 2\langle W, \mu_{\mathcal{H}_K} \rangle_{\mathcal{H}_K} + \langle W, \Sigma_{\mathcal{H}_K} W \rangle_{\mathcal{H}_K}.$$

For an arbitrary direction $V \in \mathcal{H}_K$,

$$DQ(W)[V] = 2\langle \Sigma_{\mathcal{H}_K} W - \mu_{\mathcal{H}_K}, V \rangle_{\mathcal{H}_K}.$$

The population first-order condition is consequently

$$\Sigma_{\mathcal{H}_K} W_K^\star = \mu_{\mathcal{H}_K}.$$

If the inverse is defined on $\mu_{\mathcal{H}_K}$ in the identifiable subspace, this gives the population representation in the proposition. The normal equation itself does not require a bounded inverse on all of $\mathcal{H}_K$.

The empirical penalized criterion is

$$1 - 2\langle W, \widehat{\mu}_T \rangle_{\mathcal{H}_K} + \langle W, \widehat{\Sigma}_T W \rangle_{\mathcal{H}_K} + \lambda \|W\|^2_{\mathcal{H}_K}.$$

Its derivative vanishes exactly when

$$(\widehat{\Sigma}_T + \lambda I)\widehat{W}_{\lambda,T} = \widehat{\mu}_T.$$

Since $\widehat{\Sigma}_T$ is positive and $\lambda > 0$, the operator on the left is boundedly invertible, with inverse norm at most $1/\lambda$. This yields the unique empirical policy in (5). □

A.4. Consequences of the primitive conditions

Lemma 1 (Consequences of return-tail regularity)**.** *Fix $\varepsilon \in \mathcal{E}$, and put $C_{\mathrm{sc}} = \sqrt{8K_4(1+K_4)}$. Write $\Sigma = \Sigma_{\mathcal{H}_K,\varepsilon}$.*

(i) *The feature $X_t = N^{-1}\sum_i R_{i,t+1} K(\cdot, Z_{i,t})$ is strongly measurable,*

$$R^p_{t+1}(W) = \langle W, X_t \rangle_{\mathcal{H}_K}, \qquad \mathbb{E}_\varepsilon\left[\exp\left(\frac{\|X_t\|^2_{\mathcal{H}_K}}{B_X^2} \right) \middle| \mathscr{F}_t \right] \le 2 \quad a.s.$$

In particular,

$$\mathbb{P}_\varepsilon(\|X_t\| > u \mid \mathscr{F}_t) \le 2e^{-u^2/B_X^2} \quad a.s.,$$

and $\mathbb{E}_\varepsilon\|X_t\|^{2q} \le 2q!B_X^{2q}$ *for integers* $q \ge 1$. *These are moment bounds, not almost-sure bounds.*

(ii) *The operator* Σ *is positive, self-adjoint, trace class, and compact, with*

$$\operatorname{Tr}\Sigma = \mathbb{E}_\varepsilon\|X_t\|^2 \le B_X^2 \log 2.$$

(iii) *Writing* $\mu = \mathbb{E}_\varepsilon X_t$, *the minimum-norm target satisfies* $\Sigma W_\varepsilon^\star = \mu$. *For every* $W \in \mathcal{H}_K$,

$$Q_\varepsilon(W) - Q_\varepsilon(W_\varepsilon^\star) = \|\Sigma^{1/2}(W - W_\varepsilon^\star)\|_{\mathcal{H}_K}^2.$$

(iv) *Let* $P_t^\star = R_{t+1}^p(W_\varepsilon^\star)$ *and* $q_t = \mathbb{E}_\varepsilon[(P_t^\star)^2 \mid \mathscr{F}_t]$. *Condition* (E4) *implies*

$$\mathbb{E}_\varepsilon[P_t^\star \mid \mathscr{F}_t] = q_t, \qquad 0 \le q_t \le 1, \qquad \mathbb{E}_\varepsilon[(m_{t+1,\varepsilon}^\star)^4 \mid \mathscr{F}_t] \le 8(1 + K_4).$$

(v) *The score* $m_{t+1,\varepsilon}^\star X_t$ *is a square-integrable Hilbert-valued martingale difference with respect to the economic timing:*

$$\mathbb{E}_\varepsilon[m_{t+1,\varepsilon}^\star X_t \mid \mathscr{F}_t] = 0, \qquad \mathbb{E}_\varepsilon[(m_{t+1,\varepsilon}^\star)^2 X_t \otimes X_t] \le C_{\mathrm{sc}}\Sigma.$$

Proof. By the reproducing property,

$$\|X_t\|_{\mathcal{H}_K}^2 = \frac{1}{N^2}\sum_{i,j=1}^N R_{i,t+1}R_{j,t+1}K(Z_{i,t}, Z_{j,t}) = \frac{1}{N^2}R_{t+1}'K_tR_{t+1}.$$

Hence Condition (E2) gives part (i) directly. Markov's inequality yields the conditional tail bound,

and the nonnegative power series of the exponential yields the moment bound after taking expectations. Measurability follows from the measurable kernel sections and separability. Conditional Jensen followed by ordinary expectation gives $\mathbb{E}\|X_t\|^2 \le B_X^2 \log 2$. Taking the expectation of the positive rank-one operator in trace norm proves part (ii).

Square integrability justifies differentiation of the population quadratic criterion. Its first-order condition is $\langle V, \mu - \Sigma W_\varepsilon^\star \rangle = 0$ for every $V \in \mathcal{H}_K$. Expanding around $W_\varepsilon^\star$ removes the cross term and proves part (iii). No bounded-support argument is used in these identities.

Taking $V = W_\varepsilon^\star$ in Condition (E4) yields $\mathbb{E}[P_t^\star \mid \mathscr{F}_t] = q_t$. Conditional Cauchy–Schwarz gives $q_t^2 \le q_t$, hence $0 \le q_t \le 1$. The conditional fourth-moment bound in Condition (E2) applies to $P_t^\star$, since its stock weights are $\mathscr{F}_t$-measurable. Thus

$$\mathbb{E}[(1 - P_t^\star)^4 \mid \mathscr{F}_t] \le 8\{1 + \mathbb{E}[(P_t^\star)^4 \mid \mathscr{F}_t]\} \le 8(1 + K_4 q_t^2) \le 8(1 + K_4).$$

For every $V \in \mathcal{H}_K$, the same condition and conditional Cauchy–Schwarz give

$$\begin{aligned}\mathbb{E}[(m^\star)^2 (R^p(V))^2 \mid \mathscr{F}_t] &\le \sqrt{\mathbb{E}[(m^\star)^4 \mid \mathscr{F}_t]\mathbb{E}[(R^p(V))^4 \mid \mathscr{F}_t]} \\ &\le C_{\mathrm{sc}} \mathbb{E}[(R^p(V))^2 \mid \mathscr{F}_t].\end{aligned}$$

Integration proves the operator inequality in part (v). Summing over an orthonormal basis gives $\mathbb{E}\|m^\star X_t\|^2 \le C_{\mathrm{sc}} \operatorname{Tr} \Sigma < \infty$. The conditional expectation therefore exists. Taking its inner products against a countable dense subset of $\mathcal{H}_K$ proves the martingale-difference identity from Condition (E4). □

Proposition 4 (A sufficient Gaussian return specification)**.** *Suppose that, conditionally on $\mathscr{F}_t$, R_{t+1} is Gaussian, possibly with a singular covariance, and*

$$\max_{1 \le i \le N} \mathbb{E}[R_{i,t+1}^2 \mid \mathscr{F}_t] \le V_R < \infty \quad a.s.,$$

uniformly in the economy and time. Then Condition (E2) *holds with $B_X^2 = 8\kappa_K V_R$ and $K_4 = 3$.*

Proof. For every $V \in \mathcal{H}_K$, $R^p_{t+1}(V)$ is conditionally Gaussian. A scalar $G \sim N(m, v)$ satisfies

$$\mathbb{E}G^4 = m^4 + 6m^2v + 3v^2 \le 3(m^2 + v)^2,$$

which proves (8) with $K_4 = 3$.

For the tail condition, Cauchy–Schwarz in $\mathcal{H}_K$ and $K(z, z) \le \kappa_K$ give

$$\|X_t\|^2_{\mathcal{H}_K} \le \frac{\kappa_K}{N} \sum_{i=1}^{N} R^2_{i,t+1}.$$

If $G \sim N(m, v)$ and $m^2 + v \le V_R$, then

$$\mathbb{E}e^{G^2/(8V_R)} = \left(1 - \frac{v}{4V_R}\right)^{-1/2} \exp\left(\frac{m^2}{8V_R - 2v}\right) \le \frac{2}{\sqrt{3}} e^{1/6} < 2.$$

Therefore, by convexity,

$$\begin{aligned}\mathbb{E}_\varepsilon\left[\exp\left(\frac{\|X_t\|^2}{8\kappa_K V_R}\right)\middle|\mathcal{F}_t\right] &\le \mathbb{E}_\varepsilon\left[\exp\left(\frac{1}{N}\sum_{i=1}^{N}\frac{R^2_{i,t+1}}{8V_R}\right)\middle|\mathcal{F}_t\right] \\ &\le \frac{1}{N}\sum_{i=1}^{N}\mathbb{E}_\varepsilon\left[\exp\left(\frac{R^2_{i,t+1}}{8V_R}\right)\middle|\mathcal{F}_t\right] < 2.\end{aligned}$$

Independence across stocks is unnecessary. □

The proposition is a sufficient condition, not a claim that every Gaussian mixture has uniform covariance-relative sub-Gaussian tails. Conditional Gaussianity with bounded conditional second moments is enough here. No Bernstein condition is imposed after conditioning on the entire augmented input (Z_t, R_{t+1}): at that point $m^\star$ is already known.

A.5. Exact reduction to the kernel learning problem

Lemma 2 (Induced KRR problem)**.** *For every $\varepsilon \in \mathcal{E}$, the direct estimator is isometrically equivalent to scalar KRR with constant response one. Under this identification, the statistical covariance operator is $\Sigma_{\mathcal{H}_K,\varepsilon}$ and the KRR prediction error is exactly population portfolio regret. The source order is $r = 1$. Since the induced kernel need not be uniformly bounded, Theorem 3.2 of Bignozzi et al. (2026) is not invoked verbatim; Lemma 3 supplies the replacement empirical-stability and score bounds.*

Proof. Step 1: data and payoff representation. Let

$$\mathcal{X} := \mathcal{Z}^N \times \mathbb{R}^N, \qquad \Xi_t := (Z_t, R_{t+1}).$$

For $x = ((z_1, \dots, z_N), r) \in \mathcal{X}$, put

$$\Phi(x) := \frac{1}{N} \sum_{i=1}^{N} r_i K(\cdot, z_i), \qquad f_W(x) := \frac{1}{N} \sum_{i=1}^{N} W(z_i) r_i.$$

Then $\Phi(\Xi_t) = X_t$ and $f_W(\Xi_t) = \langle W, X_t \rangle_{\mathcal{H}_K} = R^p_{t+1}(W)$. The symbol Ξ_t denotes the augmented observation; X_t continues to denote the managed-payoff feature used in the main text.

Let $\mathcal{G} := \{f_W : W \in \mathcal{H}_K\}$ and equip it with $\langle f_W, f_V \rangle_{\mathcal{G}} := \langle W, V \rangle_{\mathcal{H}_K}$. This is well defined: if f_W vanishes on $\mathcal{X}$, choose $r = (N, 0, \dots, 0)$ and vary z_1 to obtain $W(z_1) = 0$. Thus the map $\mathcal{T} : W \mapsto f_W$ is a Hilbert-space isometry onto $\mathcal{G}$. This uses the declared input domain, not a full-support assumption on the actual return law.

With $Y_t \equiv 1$, the empirical objectives agree exactly:

$$\frac{1}{T} \sum_{t=1}^{T} |Y_t - f_W(\Xi_t)|^2 + \lambda \|f_W\|^2_{\mathcal{G}} = \frac{1}{T} \sum_{t=1}^{T} (1 - R^p_{t+1}(W))^2 + \lambda \|W\|^2_{\mathcal{H}_K}.$$

This is an algebraic reformulation of the portfolio criterion, not an additional specification of the

return-generating process.

Step 2: induced kernel and population operator. The reproducing kernel of $\mathcal{G}$ is

$$\widetilde{K}(x, x') = \langle \Phi(x), \Phi(x') \rangle_{\mathcal{H}_K} = \frac{1}{N^2} \sum_{i,j} r_i s_j K(z_i, z'_j), \qquad x' = ((z'_1, \ldots, z'_N), s).$$

It is measurable and $\widetilde{K}(x, x) = \|\Phi(x)\|^2$, without a uniform bound on this diagonal. Nevertheless, $\mathbb{E}_\varepsilon \widetilde{K}(\Xi_t, \Xi_t) < \infty$ by Lemma 1, so the canonical embedding is bounded. Let $J : \mathcal{G} \to L^2(\mathbb{P}_{\varepsilon,\Xi})$ be the canonical embedding. For $W, V \in \mathcal{H}_K$,

$$\langle J\mathcal{T}W, J\mathcal{T}V \rangle_{L^2} = \mathbb{E}_\varepsilon[\langle W, X_t \rangle \langle V, X_t \rangle] = \langle W, \Sigma_{\mathcal{H}_K,\varepsilon} V \rangle_{\mathcal{H}_K}.$$

Consequently,

$$\mathcal{T}^* J^* J \mathcal{T} = \Sigma_{\mathcal{H}_K,\varepsilon}, \qquad \mathcal{T}^* J^* 1 = \mu_{\mathcal{H}_K,\varepsilon}.$$

The minimum-norm population target is $f_{\mathcal{G}} = \mathcal{T}W^\star_\varepsilon$, so $\|f_{\mathcal{G}}\|_{\mathcal{G}} \le R$. This is precisely source order $r = 1$. The positive eigenvalues and effective dimension are unchanged by the isometry.

Step 3: dependence, filtration, and statistical residual. The space $\mathcal{X}$ is Polish, and $\{(\Xi_t, Y_t)\}$ inherits stationarity and the mixing bound from Condition (E1). Define its statistical history by

$$\mathcal{A}_t := \sigma(\Xi_s, Y_s : s \le t), \qquad \mathcal{B}_t := \mathcal{A}_{t-1} \vee \sigma(\Xi_t).$$

The economic timing in Definition 1 implies $\mathcal{A}_{t-1} \subseteq \mathcal{F}_t$. The statistical residual is

$$\xi_t := Y_t - f_{\mathcal{G}}(\Xi_t) = 1 - R^p_{t+1}(W^\star_\varepsilon) = m^\star_{t+1,\varepsilon}.$$

By Lemma 1(v) and the tower property,

$$\mathbb{E}_\varepsilon[X_t \xi_t \mid \mathcal{A}_{t-1}] = 0.$$

Since $\widetilde{K}(\cdot, \Xi_t) = \mathcal{T} X_t$, the corresponding kernel-weighted residual in $\mathcal{G}$ is also a filtered innovation. Lemma 1 supplies the score variance bound. The residual ξ_t is a projection residual and is $\mathcal{B}_t$-measurable. Consequently, an almost-sure conditional Bernstein bound given $\mathcal{B}_t$ would force it to be bounded; it is not imported from the companion theorem. We also do not assume $\mathbb{E}[\xi_t \mid \Xi_t] = 0$. The exact replacement for the probabilistic step is given next.

Step 4: identify the risk. For any fixed $W \in \mathcal{H}_K$, Lemma 1(iii) gives

$$\|Jf_W - Jf_{\mathcal{G}}\|_{L^2}^2 = \mathbb{E}_\varepsilon[(R_{t+1}^p(W) - R_{t+1}^p(W_\varepsilon^\star))^2] = Q_\varepsilon(W) - Q_\varepsilon(W_\varepsilon^\star).$$

The same identity applies to an estimated policy when the training data are held fixed in the population evaluation. All constants are inherited from the fixed representation and the parameters of $\mathcal{E}$. □

A.6. Empirical stability with unbounded managed features

This subsection replaces the empirical-stability and noise proposition in Bignozzi et al. (2026). The covariance calculation uses their alternating-block/Berbee argument, with an analytical truncation of the feature norm. The score is controlled by a direct martingale second-moment argument. The estimator itself is not truncated, and the remaining operator decomposition and source-bias calculation are unchanged.

Lemma 3 (Unbounded-feature stability and score)**.** *Assume Conditions* (E1)–(E4) *and* $\operatorname{Tr}\Sigma < \infty$. *Put*

$$C_{\mathrm{sc}} := \sqrt{8K_4(1 + K_4)}.$$

For $0 < \lambda \leq 1$, *put*

$$D_\lambda = (\Sigma + \lambda I)^{-1/2}, \quad C_\lambda = D_\lambda \Sigma D_\lambda, \quad d_\lambda = \operatorname{Tr} C_\lambda = C_K(\lambda),$$

$$\Delta_{\lambda,T} = D_\lambda(\Sigma - \widehat{\Sigma}_T)D_\lambda, \qquad S_{\lambda,T} = \frac{1}{T}\sum_{t=1}^{T} D_\lambda m^\star_{t+1} X_t.$$

Fix $\delta \in (0,1)$ *and define*

$$\begin{aligned}
\ell &= \max\left\{1, \left\lceil (8C_\beta T/\delta)^{1/(\gamma+1)} \right\rceil\right\}, \\
H &= \log\left(\frac{eT}{\delta\lambda}\right), \qquad \tau^2 = 8B_X^2 H, \qquad B_\lambda = \frac{1+\tau^2}{\lambda}, \\
z &= \log\left(\frac{112(1+d_\lambda)}{\delta}\right).
\end{aligned}$$

If $\ell \le T/4$*, there is a numerical constant* C *such that, with probability at least* $1-\delta$*,*

$$\|\Delta_{\lambda,T}\|_{\mathrm{op}} \le C\left\{\sqrt{\frac{\ell B_\lambda z}{T}} + \frac{\ell B_\lambda z}{T}\right\} + \frac{2B_X^2(8H+1)}{\lambda} e^{-8H}, \tag{A.1}$$

and

$$\|S_{\lambda,T}\|^2_{\mathcal{H}_K} \le \frac{2C_{\mathrm{sc}}}{\delta}\frac{d_\lambda}{T}. \tag{A.2}$$

Consequently, for each fixed δ *there is a class-uniform* C_δ *such that, for sufficiently large* T*, the sufficient condition*

$$T^{\gamma/(\gamma+1)}\lambda \ge C_\delta \left[\log\left(\frac{C_\delta T}{\lambda}\right)\right]^2 \tag{A.3}$$

implies $\|\Delta_{\lambda,T}\|_{\mathrm{op}} \le 1/2$ *and* (A.2) *with probability at least* $1-\delta$*.*

Proof. We suppress the economy index. Only the covariance is truncated in the proof. Let

$$A_t = (D_\lambda X_t) \otimes (D_\lambda X_t)\mathbf{1}_{\{\|X_t\| \le \tau\}}, \qquad C_{\lambda,\tau} = \mathbb{E}A_t, \qquad H_t = A_t - C_{\lambda,\tau}.$$

The process (H_t) is centered and has no larger absolute-regularity coefficients than (Z_t, R_{t+1}). Since $0 \preceq C_{\lambda,\tau} \preceq C_\lambda \preceq I$ and $\|A_t\|_{\mathrm{op}} \le \tau^2/\lambda$, we have

$$\|H_t\|_{\mathrm{op}} \le B_\lambda, \qquad \mathbb{E}H_t^2 = \mathbb{E}A_t^2 - C_{\lambda,\tau}^2 \preceq B_\lambda C_\lambda. \tag{A.4}$$

The second inequality is an operator inequality; no independence of successive observations is used.

Let $n = \lfloor T/(2\ell) \rfloor$ and split the first $2n\ell$ observations into $2n$ consecutive blocks of length ℓ. Treat the odd and even blocks separately. For an odd-block sum B_j^o, operator Cauchy–Schwarz gives

$$\|B_j^o\|_{\mathrm{op}} \le \ell B_\lambda, \qquad \mathbb{E}(B_j^o)^2 \le \ell^2 B_\lambda C_\lambda.$$

Indeed, for every h, $\|\sum_{t\in I} H_t h\|^2 \le \ell \sum_{t\in I} \|H_t h\|^2$. Berbee coupling replaces each family by independent blocks with identical block laws. The total probability of a coupling failure for both families is at most

$$2n\beta(\ell) \le C_\beta T \ell^{-(\gamma+1)} \le \delta/8.$$

For the independent copies, a variance majorant has operator norm at most $n\ell^2 B_\lambda$ and trace at most $n\ell^2 B_\lambda d_\lambda$. The trace form of the self-adjoint Bernstein inequality (Minsker, 2017, proof of Theorem 3.1 and Section 3.2), with this variance majorant, yields for each family, with failure probability at most $\delta/8$,

$$\left\| \frac{1}{T} \sum_{j=1}^{n} B_j^{o,*} \right\|_{\mathrm{op}} \le C \left\{ \sqrt{\frac{\ell B_\lambda z}{T}} + \frac{\ell B_\lambda z}{T} \right\}.$$

The trace-to-variance-majorant ratio is at most d_λ; no bound on the effective rank of the actual variance is inferred by monotonicity. Apply the finite-dimensional inequality to increasing finite-rank projections and pass to the limit in operator norm, as in that reference. The operators here are trace class and bounded after truncation. The fewer than 2ℓ leftover observations contribute at most $2\ell B_\lambda/T$, already absorbed because $z \ge 1$.

It remains to compare the truncated and original covariances. Lemma 1 gives

$$\mathbb{P}\left(\max_{t\le T} \|X_t\| > \tau \right) \le 2Te^{-8H} \le \delta/8.$$

On the complementary event, the truncated empirical covariance equals the original empirical co-

variance. The population truncation bias satisfies

$$\begin{aligned}\|C_\lambda - C_{\lambda,\tau}\|_{\mathrm{op}} &\le \lambda^{-1}\mathbb{E}[\|X_t\|^2 \mathbf{1}_{\{\|X_t\|>\tau\}}] \\ &\le \frac{2}{\lambda}(\tau^2 + B_X^2)e^{-\tau^2/B_X^2} = \frac{2B_X^2(8H+1)}{\lambda}e^{-8H}.\end{aligned}$$

The tail-integration formula justifies the second line. This proves (A.1) with total failure probability at most $\delta/2$.

For the score, Lemma 1 gives

$$\mathbb{E}\|D_\lambda m_{t+1}^\star X_t\|^2 \le C_{\mathrm{sc}}\,\mathrm{Tr}(D_\lambda \Sigma D_\lambda) = C_{\mathrm{sc}} d_\lambda.$$

For $s < t$, the score at s is $\mathscr{F}_t$-measurable, whereas the conditional mean of the score at t given $\mathscr{F}_t$ is zero. All cross inner products consequently have zero expectation. Hence

$$\mathbb{E}\|S_{\lambda,T}\|^2 = \frac{1}{T^2}\sum_{t=1}^{T}\mathbb{E}\|D_\lambda m_{t+1}^\star X_t\|^2 \le C_{\mathrm{sc}}\frac{d_\lambda}{T}.$$

Markov's inequality proves (A.2) with failure probability $\delta/2$. There is no blocking loss in this score bound.

Finally, $d_\lambda \le B_X^2 \log 2/\lambda$ and, for fixed δ, $\ell \le C_\delta T^{1/(\gamma+1)}$. Thus the leading square-root term in (A.1) is bounded by a constant times

$$\left\{\frac{[\log(C_\delta T/\lambda)]^2}{T^{\gamma/(\gamma+1)}\lambda}\right\}^{1/2}.$$

The linear term is bounded by the corresponding squared expression. The truncation bias tends to zero uniformly for $0 < \lambda \le 1$ as $T \to \infty$. Increasing C_δ proves the final statement. □

Corollary 1 (Finite-sample portfolio loss)**.** *Under the assumptions of Lemma 3, for each fixed*

$\delta \in (0,1)$ *and sufficiently large* T *satisfying* (A.3)*,*

$$Q_\varepsilon(\widehat{W}_{\lambda,T}) - Q_\varepsilon(W_\varepsilon^\star) \le 9R^2\lambda + C_\delta \frac{C_K(\lambda)}{T}$$

with probability at least $1-\delta$*, uniformly over the class.*

Proof. Use the error decomposition, source-bias bound and operator-interaction bound of Bignozzi et al. (2026), specialized to $r = 1$. For completeness, write

$$\begin{aligned} W_\lambda &= (\Sigma + \lambda I)^{-1}\Sigma W^\star, \qquad B = W_\lambda - W^\star, \\ V_1 &= (\widehat{\Sigma}_T + \lambda I)^{-1}\frac{1}{T}\sum_t m_{t+1}^\star X_t, \\ V_2 &= (\widehat{\Sigma}_T + \lambda I)^{-1}(\Sigma - \widehat{\Sigma}_T)B. \end{aligned}$$

Then $\widehat{W} - W^\star = B + V_1 + V_2$. Spectral calculus gives $\|\Sigma^{1/2}B\|^2 \le R^2\lambda$ and $\|(\Sigma+\lambda I)^{1/2}B\|^2 \le 2R^2\lambda$. On the stability event,

$$\|\Sigma^{1/2}V_1\|^2 \le 4\|S_{\lambda,T}\|^2, \qquad \|\Sigma^{1/2}V_2\|^2 \le 2R^2\lambda.$$

The first inequality follows from $\|(I-\Delta_{\lambda,T})^{-1}\| \le 2$; the second follows from $\|(I-\Delta_{\lambda,T})^{-1}\Delta_{\lambda,T}\| \le 1$. Finally, $\|u+v+w\|^2 \le 3(\|u\|^2+\|v\|^2+\|w\|^2)$ and Lemma 3 give the claim. In particular, one may take the score constant $24C_{\mathrm{sc}}/\delta$. □

A.7. From portfolio loss to the Sharpe ratio

Lemma 4 (Sharpe-ratio recovery without loss of exponent)**.** *Let* $\widehat{W}_T \in \mathcal{H}_K$ *be any estimated policy satisfying, uniformly over* $\varepsilon \in \mathcal{E}$*,*

$$D_T := Q_\varepsilon(\widehat{W}_T) - Q_\varepsilon(W_\varepsilon^\star) = O_{\mathbb{P}_\varepsilon}(a_T), \qquad a_T \longrightarrow 0.$$

Then the estimated payoff has positive mean and positive variance with probability tending to one, uniformly, and

$$\mathrm{SR}^{\star}_{\varepsilon,K} - \mathrm{SR}(R^{p}_{t+1}(\widehat{W}_T)) = O_{\mathbb{P}_\varepsilon}(a_T).$$

Sharpe may be assigned any finite convention on a zero-variance event, whose probability tends to zero. No sign adjustment using unknown population moments is required.

Proof. Write $q^\star = Q_\varepsilon(W^\star_\varepsilon)$ and $S^\star = \mathrm{SR}^\star_{\varepsilon,K}$. By Proposition 2 and Condition (E6),

$$\underline{q} := \frac{1}{1+\overline{S}^2} \le q^\star \le \frac{1}{1+\underline{S}^2} =: \overline{q} < 1.$$

Step 1: recover the sign and exclude degeneracy. On the event $D_T < 1 - \overline{q}$, which has probability tending to one uniformly, $Q_\varepsilon(\widehat{W}_T) < 1$. Expanding this loss yields

$$2\mathbb{E}_\varepsilon[R^{p}_{t+1}(\widehat{W}_T)] > \mathbb{E}_\varepsilon[(R^{p}_{t+1}(\widehat{W}_T))^2] \ge 0.$$

Thus its mean is strictly positive. It cannot have zero variance: a zero-variance payoff with positive mean is a positive constant almost surely, and scaling the same policy would produce the unit payoff, contradicting $q^\star \ge \underline{q} > 0$. Hence its Sharpe ratio $\widehat{S}_T$ is positive on this event.

Step 2: bound squared-Sharpe loss. Optimizing only the scale of $\widehat{W}_T$ gives

$$q^\star \le \frac{1}{1+\widehat{S}_T^2} = \min_{a\in\mathbb{R}} Q_\varepsilon(a\widehat{W}_T) \le Q_\varepsilon(\widehat{W}_T) = q^\star + D_T.$$

Therefore

$$0 \le (S^\star)^2 - \widehat{S}_T^2 \le \frac{1}{q^\star} - \frac{1}{q^\star + D_T} = \frac{D_T}{q^\star(q^\star + D_T)} \le \frac{D_T}{\underline{q}^2}.$$

Step 3: pass to the level Sharpe. Since $S^\star + \widehat{S}_T \geq \underline{S}$ on the same event,

$$0 \leq S^\star - \widehat{S}_T = \frac{(S^\star)^2 - \widehat{S}_T^2}{S^\star + \widehat{S}_T} \leq \frac{D_T}{\underline{S}\,\underline{q}^2}.$$

This proves the stated rate uniformly over $\mathcal{E}$. The complementary event has uniformly vanishing probability. □

A.8. Proof of the portfolio-learning rate

Proof of Theorem 1. Fix $\varepsilon \in \mathcal{E}$.

Step 1: use the unbounded-feature extension. The proof follows the operator decomposition of Bignozzi et al. (2026), with its empirical-stability and noise step replaced by Lemma 3. Corollary 1 therefore gives, for every fixed $\delta \in (0,1)$, a constant C_δ common to the class such that

$$Q_\varepsilon(\widehat{W}_{\lambda,T}) - Q_\varepsilon(W_\varepsilon^\star) \leq C_\delta \left(\lambda + \frac{C_{K,\varepsilon}(\lambda)}{T}\right)$$

with probability at least $1-\delta$, for sufficiently large T satisfying

$$T^{\gamma/(\gamma+1)}\lambda \geq C_\delta[\log(C_\delta T/\lambda)]^2. \tag{A.5}$$

The extra logarithm occurs only in this sufficient stability condition, not in the leading score variance. By Condition (E5) and Proposition A.6 of Bignozzi et al. (2026), $C_{K,\varepsilon}(\lambda) \asymp \lambda^{-1/b}$ uniformly.

Step 2: verify the admissible regularization scale. For $\lambda_T^\star \asymp T^{-b/(b+1)}$, the class restriction $\gamma > b$ implies

$$\frac{\gamma}{\gamma+1} - \frac{b}{b+1} = \frac{\gamma - b}{(\gamma+1)(b+1)} > 0.$$

Consequently,

$$\frac{T^{\gamma/(\gamma+1)}\lambda_T^\star}{[\log(C_\delta T/\lambda_T^\star)]^2} \longrightarrow \infty,$$

so (A.5) holds eventually. At this scale,

$$\lambda_T^\star \asymp \frac{C_{K,\varepsilon}(\lambda_T^\star)}{T} \asymp T^{-b/(b+1)}.$$

This proves the population-loss rate. Uniformity means that, for each $\delta \in (0,1)$, some $C_\delta, T_0(\delta)$ independent of ε satisfy

$$\sup_{\varepsilon\in\mathcal{E}} \mathbb{P}_\varepsilon\Big(Q_\varepsilon(\widehat{W}_{\lambda_T^\star,T}) - Q_\varepsilon(W_\varepsilon^\star) > C_\delta T^{-b/(b+1)}\Big) \le \delta, \qquad T \ge T_0(\delta).$$

Step 3: translate loss into Sharpe and frontier recovery. Apply Lemma 4 with $a_T = T^{-b/(b+1)}$. At any fixed volatility target σ, the difference between the population and learned capital-allocation lines is

$$\mu_K^\star(\sigma) - \widehat{\mu}_{K,T}(\sigma) = \sigma\left(\mathrm{SR}_{\varepsilon,K}^\star - \mathrm{SR}(R_{t+1}^p(\widehat{W}_{\lambda_T^\star,T}))\right),$$

which has the same order, also uniformly for σ in a bounded interval. This concerns the tangency capital-allocation line with a risk-free asset and scalar rescaling, not the entire risky-only frontier.

□

A.9. Effective portfolio complexity

Proof of Theorem 2. The proof of Theorem 1 already establishes admissibility of $\lambda_T^\star$ and the spectral relation $C_{K,\varepsilon}(\lambda) \asymp \lambda^{-1/b}$. Thus

$$C_{K,T}^\star = C_{K,\varepsilon}(\lambda_T^\star) \asymp T^{1/(b+1)}, \qquad \frac{C_{K,T}^\star}{T} \asymp T^{-b/(b+1)} \longrightarrow 0.$$

It is finite for every T, since $C_{K,\varepsilon}(\lambda) \le \mathrm{Tr}(\Sigma_{\mathcal{H}_K,\varepsilon})/\lambda$.

To express the same balance in complexity units, let $C = C_{K,\varepsilon}(\lambda)$, so that $\lambda \asymp C^{-b}$. The leading

risk upper envelope has the form

$$\mathfrak{R}_T(C) = aC^{-b} + d\frac{C}{T}, \qquad a, d > 0.$$

Its derivatives are

$$\mathfrak{R}'_T(C) = -baC^{-b-1} + \frac{d}{T}, \qquad \mathfrak{R}''_T(C) = b(b+1)aC^{-b-2} > 0.$$

Hence its unique minimum occurs at

$$C_T^{\mathrm{env}} = \left(\frac{ba}{d}T\right)^{1/(b+1)}.$$

This is an upper-envelope calculation, not a shape theorem for the exact Sharpe curve of every economy. No matching pointwise bias lower bound is used. Condition (E6) is not needed for this conclusion. □

A.10. A finance-specific minimax construction

Proof of Theorem 3. The construction keeps the policy kernel fixed. It is a least-favourable family used only in this lower-bound proof, not a structural return equation imposed on the economies in the main text.

Step 1: construct a least-favourable family with nondegenerate Sharpe. Since $\mathcal{E}$ is nonempty, take the one-period state-return marginal ν_0 of one economy in the class. Write a draw as $x = (\mathbf{z}, r)$ and put

$$Q_X(x) := \|\Phi(x)\|^2_{\mathcal{H}_K}, \qquad q_X := \int Q_X(x)\, d\nu_0(x).$$

Condition (E2) and Jensen's inequality imply $q_X \le B_X^2 \log 2$. Moreover $q_X > 0$, because otherwise the managed feature would vanish almost surely, contradicting the spectral lower bound in Condition (E5).

Define a tilted law by

$$d\widetilde{\nu}(x) = \frac{Q_X(x)}{q_X}\, d\nu_0(x),$$

and, on the support of the tilted law, replace the return vector by

$$r^\circ = \sqrt{\frac{q_X}{Q_X(x)}}\, r.$$

Let ν denote the law of $(\mathbf{z}, r^\circ)$ and reuse r for the normalized return vector below. Since the feature map is linear in returns,

$$\int \Phi(\mathbf{z}, r^\circ) \otimes \Phi(\mathbf{z}, r^\circ)\, d\widetilde{\nu} = \int \Phi(x) \otimes \Phi(x)\, d\nu_0(x) =: \Sigma_0.$$

Thus the normalization preserves the entire managed-payoff second-moment operator. Under ν,

$$\|\Phi(x)\|^2_{\mathcal{H}_K} = q_X \le B_X^2 \log 2 \quad \text{almost surely.}$$

Put $\overline{B}_X := B_X\sqrt{\log 2}$.

Write

$$\Sigma_0 e_j = \mu_j e_j, \qquad c_\mu j^{-b} \le \mu_j \le C_\mu j^{-b}.$$

Its eigenvectors are orthonormal in $\mathcal{H}_K$. Fix constants $a, h > 0$, with $h \le a$, sufficiently small that

$$a^2 + h^2 \le R^2, \qquad \overline{B}_X\sqrt{a^2 + h^2} \le \frac{1}{2}, \qquad 8C_\mu h^2 \le \frac{1}{8}.$$

For an integer $M \ge 1$ and $\theta \in \{-1, 1\}^M$, define

$$W_\theta := ae_1 + \frac{h}{\sqrt{M}} \sum_{j=M+1}^{2M} \theta_{j-M} e_j, \qquad g_\theta(x) := \langle W_\theta, \Phi(x)\rangle_{\mathcal{H}_K}.$$

Then

$$\|W_\theta\|_{\mathcal{H}_K} = \sqrt{a^2 + h^2} \le R, \qquad |g_\theta(x)| \le \overline{B}_X \sqrt{a^2 + h^2} \le \frac{1}{2}.$$

Put

$$A_M := \|g_\theta\|^2_{L^2(\nu)} = a^2 \mu_1 + \frac{h^2}{M} \sum_{j=M+1}^{2M} \mu_j.$$

This quantity does not depend on θ. Moreover,

$$a^2 c_\mu \le A_M \le \frac{1}{4}.$$

Step 2: construct admissible financial observations. Draw $x_t = (Z_t, r_t)$ independently with law ν. Conditional on x_t, draw $s_t \in \{-1, 1\}$ according to

$$\mathbb{P}_\theta(s_t = 1 \mid x_t) = \frac{1 + g_\theta(x_t)}{2}.$$

The observed financial row is $(Z_t, R_{t+1}) = (Z_t, s_t r_t)$. For the least-favourable construction take

$$\mathscr{F}_t = \sigma(x_u, s_u : u < t) \vee \sigma(x_t).$$

This information structure is allowed by Definition 1. Rows are i.i.d. The managed feature is $s_t \Phi(x_t)$ and, since $s_t^2 = 1$,

$$\|s_t \Phi(x_t)\|^2_{\mathcal{H}_K} = q_X.$$

Because x_t is contained in $\mathscr{F}_t$,

$$\mathbb{E}_\theta \left[\exp\left(\frac{\|s_t \Phi(x_t)\|^2_{\mathcal{H}_K}}{B_X^2} \right) \middle| \mathscr{F}_t \right] = \exp(q_X / B_X^2) \le 2.$$

Moreover, for every $V \in \mathcal{H}_K$,

$$(R^p_{t+1}(V))^2 = \langle V, \Phi(x_t)\rangle^2_{\mathcal{H}_K}, \qquad (R^p_{t+1}(V))^4 = \langle V, \Phi(x_t)\rangle^4_{\mathcal{H}_K}$$

conditionally on $\mathscr{F}_t$. Hence Condition (E2) holds with fourth-moment coefficient one, and therefore with the class constant $K_4 \geq 1$. The second-moment operator of the managed feature is exactly Σ_0.

For every $V \in \mathcal{H}_K$, conditional on the current x_t and the past,

$$\begin{aligned}&\mathbb{E}_\theta\big[(1 - R^p_{t+1}(W_\theta))R^p_{t+1}(V) \mid x_t, \mathscr{F}_{t-1}\big] \\ &\qquad = \langle V, \Phi(x_t)\rangle_{\mathcal{H}_K}\mathbb{E}_\theta[s_t - g_\theta(x_t) \mid x_t, \mathscr{F}_{t-1}] = 0.\end{aligned}$$

Thus Condition (E4) holds. For any portfolio policy W with finite loss, write $g_W(x) = N^{-1}\sum_i W(z_i)r_i$. Then

$$Q_\theta(W) - Q_\theta(W_\theta) = \int (g_W(x) - g_\theta(x))^2\, d\nu(x). \tag{A.6}$$

Indeed, conditioning on x gives the quadratic $1 - 2g_\theta(x)g_W(x) + g_W(x)^2$. Consequently, W_θ is the population-optimal minimum-norm policy.

Its payoff satisfies

$$\mathbb{E}_\theta[R^p_{t+1}(W_\theta)] = A_M, \qquad \mathbb{E}_\theta[(R^p_{t+1}(W_\theta))^2] = A_M.$$

Hence

$$(\mathrm{SR}^\star_\theta)^2 = \frac{A_M}{1 - A_M}.$$

Because A_M is uniformly bounded away from zero and from one, the optimal Sharpe ratio is uniformly bounded away from zero and infinity. Thus the least-favourable family satisfies Condition (E6), with constants common to the family, as well as Conditions (E1)–(E5).

Step 3: neighboring economies are difficult to distinguish. Consider the more informative experiment that also reveals x_t and s_t. A lower bound in this augmented experiment also applies to the

financial observations. Neighboring parameters differ in one sign, say coordinate $j - M$. Their conditional Bernoulli probabilities lie in $[1/4, 3/4]$, and

$$\mathrm{KL}(\mathrm{Ber}(p), \mathrm{Ber}(q)) \le \frac{(p-q)^2}{q(1-q)}.$$

Hence, for the laws $\mathbb{P}_\theta^{\mathrm{aug},T}$ of T augmented observations and a one-bit neighbor θ', we have

$$\begin{aligned}\mathrm{KL}(\mathbb{P}_\theta^{\mathrm{aug},T}, \mathbb{P}_{\theta'}^{\mathrm{aug},T}) &\le 2T \int (g_\theta - g_{\theta'})^2 \, d\nu \\ &= \frac{8Th^2\mu_j}{M} \\ &\le 8C_\mu h^2 \frac{T}{M^{b+1}}.\end{aligned}$$

Choose

$$M = \left\lceil T^{1/(b+1)} \right\rceil.$$

The right-hand side is at most $1/8$, so Pinsker's inequality gives total variation at most $1/4$ for every neighboring pair.

Step 4: testing difficulty implies scale-free directional error. Define

$$\psi_j(x) := \frac{\langle e_j, \Phi(x) \rangle_{\mathcal{H}_K}}{\sqrt{\mu_j}},$$

so that $(\psi_j)_{j \ge 1}$ is orthonormal in $L^2(\nu)$. Let $\widehat{g}$ be any estimated square-integrable payoff rule and write

$$\widehat{a}_j := \langle \widehat{g}, \psi_j \rangle_{L^2(\nu)}.$$

Decode the signs by

$$\widehat{\theta}_{j-M} := \mathrm{sign}(\widehat{a}_1 \widehat{a}_j), \qquad M < j \le 2M,$$

with any fixed convention at zero. This decoder is invariant to any nonzero global rescaling of $\widehat{g}$.

Define the scale-free directional error

$$D_\theta(\widehat{g}) := \inf_{c\in\mathbb{R}} \|g_\theta - c\widehat{g}\|^2_{L^2(\nu)}.$$

For any $c \in \mathbb{R}$, if $c\widehat{a}_1 \le 0$, the squared error in the first coordinate is at least $a^2\mu_1$. If $c\widehat{a}_1 > 0$, every incorrectly decoded sign contributes at least $h^2\mu_j/M$ to the squared error. Since $h \le a$ and $H(\widehat{\theta},\theta) \le M$, both cases imply

$$D_\theta(\widehat{g}) \ge \frac{h^2 c_\mu}{2^b M^{b+1}} H(\widehat{\theta},\theta).$$

For every neighboring pair, the sum of the two sign-error probabilities is at least $1 - \mathrm{TV} \ge 3/4$. Averaging over the hypercube and summing over its M coordinates therefore gives

$$2^{-M}\sum_\theta \mathbb{E}_\theta[H(\widehat{\theta},\theta)] \ge \frac{3M}{8}.$$

Consequently,

$$2^{-M}\sum_\theta \mathbb{E}_\theta[D_\theta(\widehat{g})] \ge \frac{3h^2 c_\mu}{2^{b+3}} M^{-b}.$$

Step 5: directional error is a Sharpe-ratio gap. Write

$$m := \langle g_\theta, \widehat{g}\rangle_{L^2(\nu)}, \qquad v := \|\widehat{g}\|^2_{L^2(\nu)}.$$

If $v > 0$, the payoff generated by $\widehat{g}$ satisfies

$$\mathbb{E}_\theta[R(\widehat{g})] = m, \qquad \mathbb{E}_\theta[R(\widehat{g})^2] = v,$$

and hence

$$\mathrm{SR}_\theta(\widehat{g})^2 = \frac{m^2}{v - m^2}.$$

Moreover,

$$D_\theta(\widehat{g}) = A_M - \frac{m^2}{v} = \frac{1}{1+\mathrm{SR}_\theta(\widehat{g})^2} - \frac{1}{1+(\mathrm{SR}^\star_\theta)^2}.$$

The same identity holds for $v = 0$ under the convention $\mathrm{SR}_\theta(\widehat{g}) = 0$. Cauchy–Schwarz implies $|\mathrm{SR}_\theta(\widehat{g})| \le \mathrm{SR}^\star_\theta$.

If $\mathrm{SR}_\theta(\widehat{g}) \ge 0$, then

$$\mathrm{SR}^\star_\theta - \mathrm{SR}_\theta(\widehat{g}) \ge \frac{D_\theta(\widehat{g})}{2\overline{S}}.$$

If $\mathrm{SR}_\theta(\widehat{g}) < 0$, then the gap is at least $\underline{S}$; since $D_\theta(\widehat{g}) \le A_M \le 1/4$, it is again bounded below by a fixed positive multiple of $D_\theta(\widehat{g})$. Therefore there is a constant $c_S > 0$, independent of M, T, θ, such that

$$\mathrm{SR}^\star_\theta - \mathrm{SR}_\theta(\widehat{g}) \ge c_S D_\theta(\widehat{g}).$$

Combining this inequality with Step 4 yields

$$2^{-M} \sum_\theta \mathbb{E}_\theta\big[\mathrm{SR}^\star_\theta - \mathrm{SR}_\theta(\widehat{g})\big] \ge c_1 M^{-b}$$

for a constant $c_1 > 0$. Since $M = \lceil T^{1/(b+1)} \rceil \le 2T^{1/(b+1)}$ for all sufficiently large T, we obtain

$$\sup_\theta \mathbb{E}_\theta\big[\mathrm{SR}^\star_\theta - \mathrm{SR}_\theta(\widehat{g})\big] \ge c_2 T^{-b/(b+1)}$$

for some $c_2 > 0$.

The argument allowed $\widehat{g}$ to range over all square-integrable estimators in the more informative augmented experiment, a larger class than managed portfolio rules based on the observed financial rows. Taking the infimum over all measurable portfolio-learning procedures proves

$$\inf_{\widetilde{W}_T} \sup_{\varepsilon \in \mathcal{E}} \mathbb{E}_\varepsilon \left[\mathrm{SR}^\star_{\varepsilon,K} - \mathrm{SR}_\varepsilon\Big(R^p_{t+1}(\widetilde{W}_T)\Big)\right] \ge c\, T^{-b/(b+1)}$$

for a constant $c > 0$ and all sufficiently large T. □

B. Tabelle

Table III. Portfolio representations generated by different kernels

Representation	Kernel	Portfolio policy
Common exposure	$K(z, z') = 1$	$W(z)$ is constant
Linear	$K(z, z') = 1 + z'z'$	$W(z)$ is linear in characteristics
Finite features	$K(z, z') = \Phi(z)'\Phi(z')$	$W(z)$ lies in the span of the chosen features
Polynomial	$K(z, z') = (1 + z'z')^q$	$W(z)$ includes polynomial terms and interactions
Gaussian	$K(z, z') = \exp\left(-\frac{\|z - z'\|^2}{2\ell^2}\right)$	Flexible nonlinear $W(z)$
Matérn	$K_s(z, z') = c_{s,D}\, \rho_s(z, z')^{s-D/2} \times \mathcal{K}_{s-D/2}\big(\rho_s(z, z')\big)$	Flexible nonlinear $W(z)$ with smoothness controlled by s
Neural tangent	$K_{\mathrm{NTK}}(z, z') = \frac{\|z\|\,\|z'\|}{2\pi}\Big[\sin\theta + 2(\pi - \theta)\cos\theta\Big]$	Infinite-width one-hidden-layer ReLU policy

Notes: Here $\ell > 0$ denotes the kernel length scale. For the Matérn kernel,

$$\rho_s(z, z') = \sqrt{2s - D}\,\frac{\|z - z'\|}{\ell}, \qquad c_{s,D} = \frac{2^{1-(s-D/2)}}{\Gamma(s - D/2)}, \qquad s > \frac{D}{2},$$

and $\mathcal{K}_\alpha$ denotes the modified Bessel function of the second kind. Under standard regularity conditions, the Matérn RKHS is equivalent to a Sobolev space of smoothness s.
For the neural tangent kernel,

$$\theta = \arccos\left(\frac{z'z'}{\|z\|\,\|z'\|}\right),$$

and the displayed expression corresponds to a one-hidden-layer ReLU network without bias in the infinite-width NTK limit, up to the conventional overall normalization.
The kernel determines the class of characteristic-to-exposure mappings available to the portfolio manager. Portfolio returns use the common cross-sectional normalization in Equation (1). The common-exposure specification is an equal-exposure characteristic rule, not an unrestricted vector of static stock weights.

Table IV relates the learning exponent to familiar spectral benchmarks. These are conditions on the managed-portfolio operator, not conclusions implied by the kernel name alone.

Table IV. Benchmark portfolio-learning rates

Portfolio class	Managed-portfolio spectrum	Sharpe learning rate
Finite dimensional	Fixed finite rank	T^{-1}
Gaussian benchmark	Fast decay	Nearly T^{-1}
Sobolev–Matérn benchmark	$b = 2\widetilde{s}/D$	$T^{-2\widetilde{s}/(2\widetilde{s}+D)}$

Notes: Sharpe rates use Condition (E6). The finite-rank rate uses a fixed identifiable payoff space and root-T moment estimation. The fast-decay row is a separate i.i.d. or sufficiently fast-mixing benchmark, with possible logarithmic factors and an admissible regularization sequence; it does not follow by setting $b = \infty$ in Theorem 1. The Sobolev–Matérn row assumes the displayed *managed-payoff* spectral law and $\gamma > b$.

C. Simulation diagnostics and numerical scope

All diagnostics added during closeout are analyses after observing the archived experiment. The economic specification, folds, grid, pairing, initial calibration time, exponent and all original outcomes are immutable. Full per-history data, normalized outcomes, distribution summaries and reproduction commands accompany the generated tables.

Table V. Initial-CV descriptive baseline slopes on the established windows. The slope is the regression of log Monte Carlo mean, not the mean of per-history slopes. Full covariance preserves complete history dependence across horizons.

Quantity	Window	Horizons	Slope	MCSE	Jackknife
Regret	full	13	-0.43782	0.00556	0.00557
Regret	T_ge_360	8	-0.40437	0.00541	0.00542
Regret	T_ge_720	6	-0.38727	0.00636	0.00637
Regret	T_ge_1440	4	-0.38125	0.00740	0.00742
Sharpe gap	full	13	-0.39865	0.00552	0.00553
Sharpe gap	T_ge_360	8	-0.37208	0.00486	0.00486
Sharpe gap	T_ge_720	6	-0.36043	0.00585	0.00586
Sharpe gap	T_ge_1440	4	-0.35854	0.00706	0.00707

Table VI. Initial-CV endpoints in the five economies, all 300 histories. The common comparison grid ends at $T = 3240$; baseline continues to 4860.

Economy	T	Annual SR	Regret
N1200	3240	2.97420	0.00877
N300	3240	2.81517	0.02529
baseline	4860	2.93250	0.01130
rho000	3240	2.91578	0.01538
rho075	3240	2.91842	0.01463

For each policy, let $u = \mathbb{E}Y$, $v = \mathbb{E}Y^2$ and $q = 1 - Q(W^\star)$. These are monthly moments. Set $c = u/q$, $d = v - u^2/q$. Direct calculation verifies

$$Q(W) - Q(W^\star) = q(c-1)^2 + d, \qquad \mathrm{SR}(W) = \frac{u}{\sqrt{v - u^2}}.$$

For $c > 0$, the scale-invariant directional error is $d/[c^2 q(1-q)]$. For $u, v > 0$, the positive-ray identity is

$$\underbrace{q - u^2/v}_{\text{ray gap}} + \underbrace{(v-u)^2/v}_{\text{scale gap}} = Q(W) - Q(W^\star).$$

Null or negative means and zero variances are recorded separately, without changing policy orientation. Roundoff-sized negative directional errors are retained; larger negative errors fail verification. Nonlinear identities are checked per policy rather than on aggregate moments.

For the same numerical subspace and actual penalty, let $\beta_P = S^{-1}m$, $\beta_\lambda = (S+\lambda I)^{-1}m$, and $\Delta = \widehat{\beta} - \beta_\lambda$. Then

$$\begin{aligned} Q(\widehat{\beta}) - Q(W^\star) = q - m'S^{-1}m + (\beta_\lambda - \beta_P)'S(\beta_\lambda - \beta_P) \\ + \Delta' S \Delta + 2\Delta' S(\beta_\lambda - \beta_P). \end{aligned}$$

The signed cross term is preserved. The displacement norm is not automatically a variance, particularly with a data-selected penalty. Recovered-moment tables explicitly state the number of histories and horizons re-estimated; they must not be read as decompositions of all 1500 archived histories.

Table VII. Baseline endpoint financial components, all 300 histories. Ray/scale quantities are uniquely identified from positive signed Sharpe and regret for each policy. Population-ridge bias uses each actual penalty in the same numerical operator. The last column keeps displacement and signed cross combined; it is not variance. Their separate values require recovered coefficients.

Specification	Ray	Scale	Floor	Bias	Displacement + cross
Initial CV	0.01112	0.00018	0.00178	0.00318	0.00634
Historical	0.01073	0.01147	0.00178	0.01934	0.00109

The numerical protocol separates A (the archived rank-4096 constant held fixed at both ranks) from B (initial CV repeated only on the identical first 60 observations at each rank). It declares twelve representative original production indices and separate stress indices selected by the smallest initial constant. These differ from the historical rank-pilot seeds. Full and short audits have separate

Table VIII. Actual recovered monthly moments and radial/angular components, baseline T=4860. Scope is the same explicitly counted audit histories as the following table; these values are not inferred by choosing roots of Q/SR.

Policy	u	v	c	d	Radial	Normalized direction	Monthly SR
A	0.429376	0.440028	1.001877	0.009846	0.000002	0.040055	0.849185
C	0.362138	0.315970	0.844989	0.009968	0.010298	0.057004	0.842350

Table IX. Recovered baseline endpoint moments. A uses frozen initial CV; C uses the historical constant. Scope: A: 1 histories, C: 1 histories; descriptive paired audit histories only. Ray plus scale and floor plus bias plus displacement plus signed cross reconstruct regret.

Policy	Ray	Scale	Floor	Bias	Displacement	Cross
A	0.00959	0.00026	0.00178	0.00368	0.00437	0.00002
C	0.01352	0.00675	0.00178	0.01934	0.00116	-0.00201

labels. The original 5% projection-floor criterion, paired absolute mean plus 1.96 MCSE relative to mean rank-4096 regret, and independent-quadrature comparison remain unchanged. Mean, second moment, Sharpe-gap and normalized-directional sensitivity thresholds are also 5%. No subset result certifies an untested horizon or provides a forward certificate for rank 8192.

The new CV audit re-estimates full audit horizons for 1 original production histories and only T=60 for 15 additional histories. There are 0 failed numerical metric comparisons and 450 comparisons with insufficient independent histories for MCSE. Rank-specific initial CV changes 8 of 80 checked selections. Representative-history MCSE is computed separately from minimum-a stress diagnostics; stress cases are never pooled into an IID MCSE. The twelve-history full-horizon protocol and separate stress coverage are incomplete; uncertainty at untested cells remains unresolved. Rank 8192 has no forward-rank certificate. Population-ridge decomposition at rank 8192 has not run. Quadrature differences are systematic evaluation diagnostics; replication MCSE does not quantify uncertainty over quadrature seeds.

The infinite Fourier tail satisfies $\|\psi_\infty - \psi_M\|_\infty \le c_\psi/[4M^4\log(M+1)]$, about 6.6991×10^{-9} at $M = 128$. For fixed policies this implies uniform loading and moment perturbation bounds. The

Table X. Paired rank8192 minus4096 endpoint changes on the recovered audit histories. A holds the archived constant fixed; B reselects once on the initial60 at each rank. These are point differences. Where n=1 a replication MCSE is not estimable; the complete representative/stress endpoint audit remains unresolved.

Economy	T	Audit	n	ΔQ	Δu	Δv	Δ SR gap	Δ direction
N1200	3240	A	1	3.669e-06	1.660e-05	3.688e-05	1.803e-05	1.235e-05
N1200	3240	B	1	3.669e-06	1.660e-05	3.688e-05	1.803e-05	1.235e-05
N300	3240	A	1	4.284e-05	7.079e-05	1.844e-04	2.051e-04	1.542e-04
N300	3240	B	1	4.284e-05	7.079e-05	1.844e-04	2.051e-04	1.542e-04
baseline	4860	A	1	2.468e-06	2.421e-05	5.089e-05	7.278e-06	5.146e-06
baseline	4860	B	1	2.468e-06	2.421e-05	5.089e-05	7.278e-06	5.146e-06
rho000	3240	A	1	-1.186e-06	6.078e-05	1.204e-04	-8.762e-06	-6.178e-06
rho000	3240	B	1	-1.186e-06	6.078e-05	1.204e-04	-8.762e-06	-6.178e-06
rho075	3240	A	1	6.156e-06	4.861e-05	1.034e-04	2.956e-05	2.096e-05
rho075	3240	B	1	6.156e-06	4.861e-05	1.034e-04	2.956e-05	2.096e-05

exported operator perturbation ratios use every actually selected penalty. Small pointwise error alone does not establish stability of a ridge inverse or an initial argmin; paired Fourier diagnostics and any remaining gaps are explicitly separated. The finite truncation does not identify a maximal source order from its differentiability.

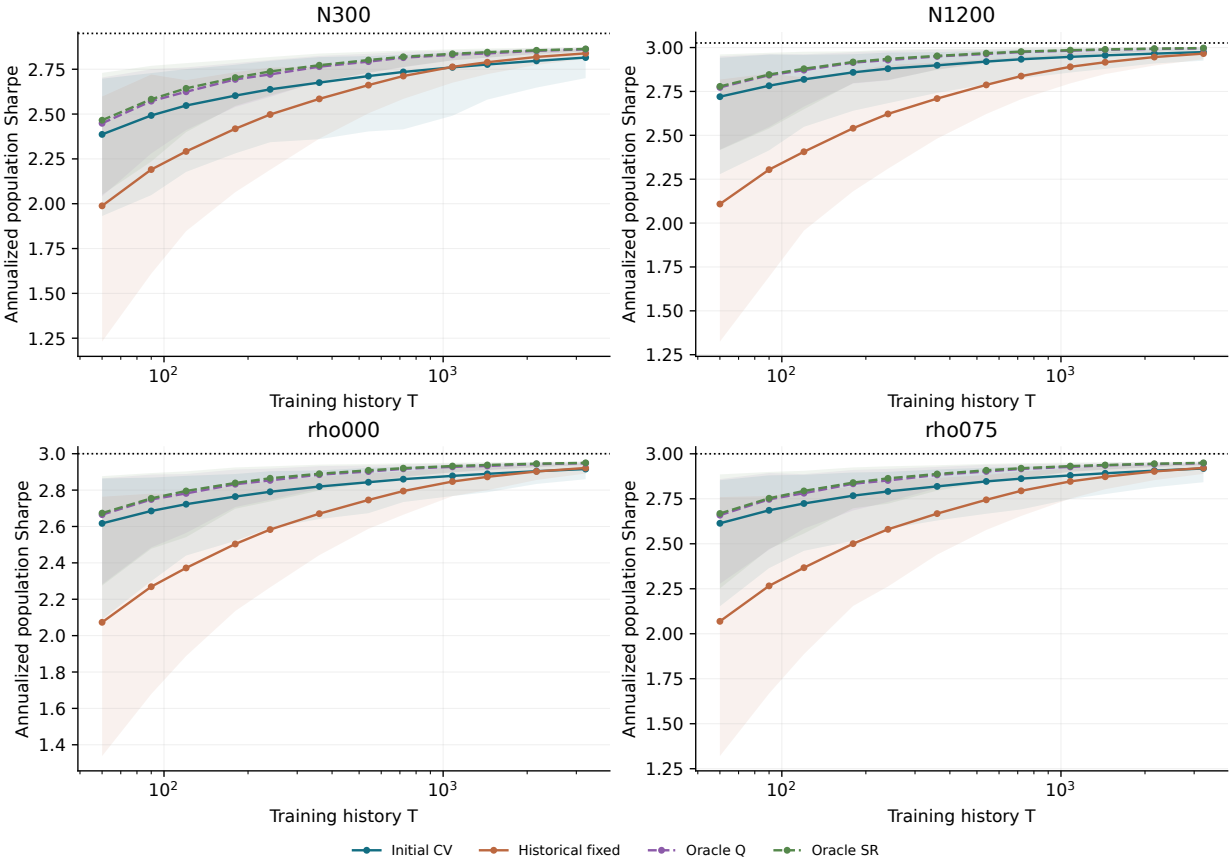


Figure 16. All four alternative economies on the common grid, with their analytic economic optima. Bands show replication outcomes. Economies differing only in persistence share the marginal population operator but can select different penalties and complexities.

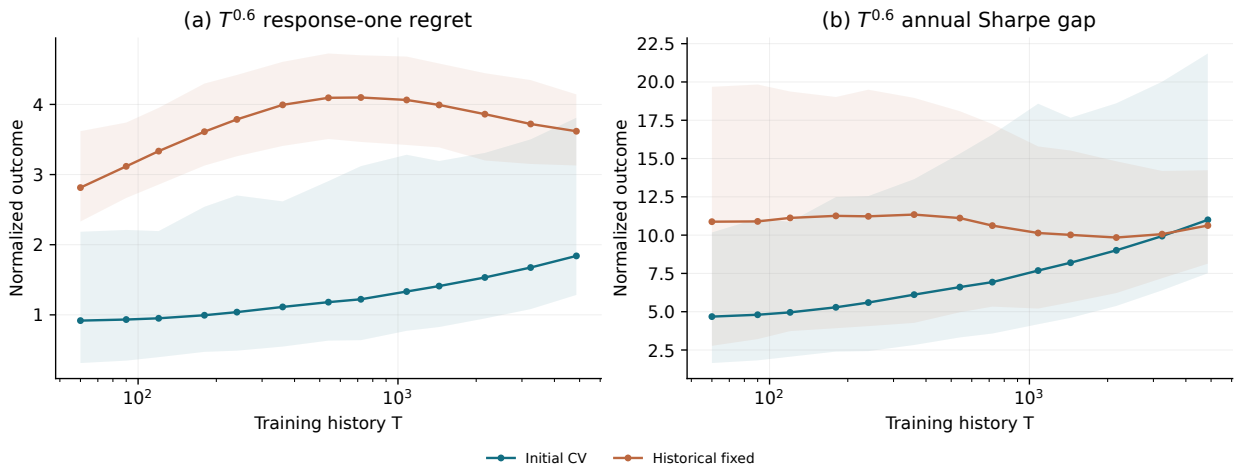


Figure 17. Normalized baseline regret and Sharpe gap on the complete grid. Means and replication-percentile bands are finite descriptive distributions; they do not identify an asymptotic rate in probability.